



SCTR's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE –
411043

(An Autonomous Institute affiliated to Savitribai Phule Pune University)
DEPARTMENT OF ELECTRONICS & COMPUTER ENGG.

Artificial Intelligence (AI) (MDM3) **(2605MD19)**

Third Year of E&C Engineering [2024 course]

Subject Teacher
Dr. A. A. Yadav

Course Objectives and Outcomes

Course Objectives:

The main objectives of this course are:

- Foundational understanding of concepts and techniques in Artificial Intelligence
- Analysing skills of various AI algorithms by comparing accuracy, efficiency, and applicability across domains
- Applying NLP modelling techniques to develop real time application
- Categorizing morphological and syntactic analysis for NLP models

Course Outcomes: After completing this course, students will be able to

- CO1: Explain fundamental principles of AI, intelligent agents, and basic search algorithms
- CO2: Design adversarial search strategies and constraint satisfaction solutions
- CO3: Perform morphological and syntactic analysis of NLP models
- CO4: Construct NLP modeling to develop real time application

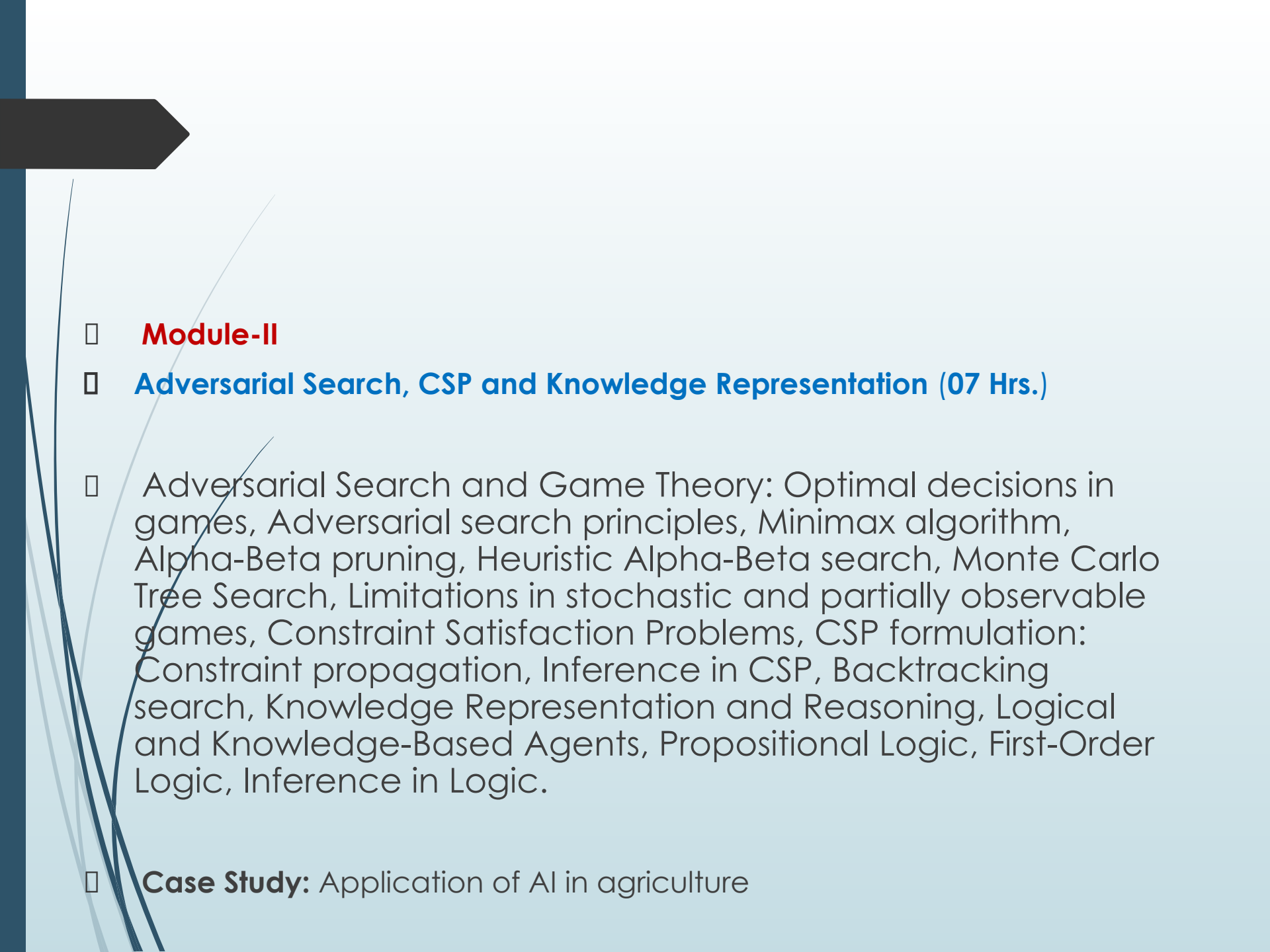


COURSE CONTENTS

- **Module-I**

- **Foundations of AI, Problem Solving and Search (06 Hrs.)**

- Introduction to Artificial Intelligence: Definitions, foundations, history, state of the art, risks and benefits of AI; Intelligent Agents: Types, structure, environments, rationality, agent behavior, nature of environments; Problem-Solving: Problem characteristics, production systems, control strategies; Search Techniques: Uninformed search (breadth-first, depth-first), informed/heuristic search (hill climbing, best first), heuristic functions.
- **Case Study:** Application of AI in Healthcare Engineering



□ **Module-II**

□ **Adversarial Search, CSP and Knowledge Representation (07 Hrs.)**

- Adversarial Search and Game Theory: Optimal decisions in games, Adversarial search principles, Minimax algorithm, Alpha-Beta pruning, Heuristic Alpha-Beta search, Monte Carlo Tree Search, Limitations in stochastic and partially observable games, Constraint Satisfaction Problems, CSP formulation: Constraint propagation, Inference in CSP, Backtracking search, Knowledge Representation and Reasoning, Logical and Knowledge-Based Agents, Propositional Logic, First-Order Logic, Inference in Logic.

- **Case Study:** Application of AI in agriculture



❑ **Module-III**

❑ **Introduction, Text Processing and Morphology (09 Hrs.)**

❑ Introduction and Core Concepts of NLP, History and Scope of NLP, Scientific vs. Engineering Goals in NLP Components of NLP, NLP Pipeline and its stages, Programming Languages vs. Natural Languages, Challenges in NLP, Problems in Natural Language Understanding, Applications of NLP, Empirical Laws: Zipf's Law, Heap's Law, Challenges in Processing Indian Languages, Text Processing and Normalization, Text Segmentation, Word Segmentation, Sentence Segmentation, Tokenization, Text Normalization, Case Folding, Removal of Punctuation, Stemming (Porter's Algorithm), Lemmatization, Spelling Correction using Edit Distance (Levenshtein Distance), Morphological Analysis, Morphemes, Inflectional vs. Derivational Morphology, Morphological Parsing, Finite-State Transducers (FST) for Morphology.

❑ **Case Study:** NLP Libraries: NLTK, spaCy, WordNet



□ **Module-IV**

□ **Language Modelling and Tagging (07 Hrs.)**

- Syntax and Parsing, Context-Free Grammar (CFG), Constituency Parsing, Probabilistic CFG (PCFG), CKY Algorithm, Dependency Parsing, Transition-Based Parsing, Graph-Based Parsing, Parsing Issues and Structural Ambiguity, Evaluation Metrics for Parsing, Language Modeling, Statistical Models of Language,
- N-gram Models, Smoothing Techniques, Additive / Laplace, Good-Turing, Class-based Models, Brown Clustering, Neural Language Models, Part-of-Speech (POS) Tagging, Rule-Based Tagging, Transformation-Based Tagging (TBT), Statistical Tagging, Hidden Markov Models (HMMs), Viterbi Algorithm, Baum-Welch Algorithm, Discriminative Models, Maximum Entropy Models, Conditional Random Fields (CRFs)



Society for Computer Technology & Research's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY
Multidisciplinary Minor Courses

Third Year B-Tech (TY B-Tech) AY (2026-27)
Electronics and Computer Engineering (E&CE)

[2605MD19]: Artificial Intelligence (AI)

Semester	Credits	Teaching Scheme	Examination Scheme
5	2	Th: 2 Hrs. / Week	ISE: 20 Marks CIE: 20 Marks ESE: 60 Marks

Prerequisite: Students should have prior knowledge of

- Statistics, programming languages
- Machine Learning and algorithms

Course Objectives: The objective of this course is to provide students with

- Foundational understanding of concepts and techniques in Artificial Intelligence
- Analyzing skills of various AI algorithms by comparing accuracy, efficiency, and applicability across domains.
- Applying NLP modeling techniques to develop real time application
- Categorizing morphological and syntactic analysis for NLP models

Course Outcomes: After completing this course, students will be able to

- C01:** Explain fundamental principles of AI, intelligent agents, and basic search algorithms
C02: Design adversarial search strategies and constraint satisfaction solutions
C03: Perform morphological and syntactic analysis of NLP models
C04: Construct NLP modeling to develop real time application

COURSE CONTENTS

Module-I

Foundations of AI, Problem Solving and Search

06 Hrs.

Introduction to Artificial Intelligence: Definitions, foundations, history, state of the art, risks and benefits of AI; Intelligent Agents: Types, structure, environments, rationality, agent behavior, nature of environments; Problem-Solving: Problem characteristics, production systems, control strategies; Search Techniques: Uninformed search (breadth-first, depth-first), informed/heuristic search (hill climbing, best first), heuristic functions.

Case Study: Application of AI in Healthcare Engineering

Module-II

Adversarial Search, CSP and Knowledge Representation

07 Hrs.

Adversarial Search and Game Theory: Optimal decisions in games, Adversarial search principles, Minimax algorithm, Alpha-Beta pruning, Heuristic Alpha-Beta search, Monte Carlo Tree Search, Limitations in stochastic and partially observable games, Constraint Satisfaction Problems, CSP formulation: Constraint propagation, Inference in CSP, Backtracking search, Knowledge Representation and Reasoning, Logical and Knowledge-Based Agents, Propositional Logic, First-Order Logic, Inference in Logic.

Case Study: Application of AI in agriculture

Module-III	Introduction, Text Processing and Morphology	09 Hrs.
Introduction and Core Concepts of NLP, History and Scope of NLP, Scientific vs. Engineering Goals in NLP Components of NLP, NLP Pipeline and its stages, Programming Languages vs. Natural Languages, Challenges in NLP, Problems in Natural Language Understanding, Applications of NLP, Empirical Laws: Zipf's Law, Heap's Law, Challenges in Processing Indian Languages, Text Processing and Normalization, Text Segmentation, Word Segmentation, Sentence Segmentation, Tokenization, Text Normalization, Case Folding, Removal of Punctuation, Stemming (Porter's Algorithm), Lemmatization, Spelling Correction using Edit Distance (Levenshtein Distance), Morphological Analysis, Morphemes, Inflectional vs. Derivational Morphology, Morphological Parsing, Finite-State Transducers (FST) for Morphology.		
Case Study: NLP Libraries: NLTK, spaCy, WordNet		
Module-IV	Language Modelling and Tagging	07 Hrs.
Syntax and Parsing, Context-Free Grammar (CFG), Constituency Parsing, Probabilistic CFG (PCFG), CKY Algorithm, Dependency Parsing, Transition-Based Parsing, Graph-Based Parsing, Parsing Issues and Structural Ambiguity, Evaluation Metrics for Parsing, Language Modeling, Statistical Models of Language, N-gram Models, Smoothing Techniques, Additive / Laplace, Good-Turing, Class-based Models, Brown Clustering, Neural Language Models, Part-of-Speech (POS) Tagging, Rule-Based Tagging, Transformation- Based Tagging (TBT), Statistical Tagging, Hidden Markov Models (HMMs), Viterbi Algorithm, Baum- Welch Algorithm, Discriminative Models, Maximum Entropy Models, Conditional Random Fields (CRFs)		
Case Study: Students will prepare a report on study of IEEE/Springer journal paper on NLP based Diagnostic Text Mining.		

Text Books:

- T1.** S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, NJ, USA: Pearson, ISBN-13: 9780134610993, 4th edition Hoboken
- T2.** J. Perkins, *Python 3 Text Processing with NLTK 3 Cookbook*, Birmingham, U.K.: Packt Publishing Ltd. ISBN-13: 9781782167853, 4th edition

Reference Books:

- R1.** S. Kaushik, *Artificial Intelligence*, India: Cengage Learning India, ISBN-13: 978-8131510995, 1st edition
- R2.** D. Jurafsky and J. H. Martin, *Speech and Language Processing*, Pearson, ISBN-13: 978-9332518414, 3rd edition.
- R3.** Jacob Eisenstein, *Natural Language Processing*, MIT Press, ISBN-10: 0262042843, 1st edition

Relevant MOOCs Course (Course name and Weblink)

1. Artificial Intelligence search methods for problem solving By Prof. Deepak Khemani | IIT Madras Artificial Intelligence Search Methods For Problem Solving–Course, https://onlinecourses.nptel.ac.in/noc21_cs79/preview
2. An Introduction to Artificial Intelligence By Prof. Mausam | IIT Delhi An Introduction to Artificial Intelligence – Course, https://onlinecourses.nptel.ac.in/noc20_cs42/preview
3. Natural Language Toolkit: <https://www.nltk.org>



Society for Computer Technology & Research's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY
Multidisciplinary Minor Courses

NPTEL Course on Natural Language Processing by Prof. Pushpak Bhattacharyya, IIT Bombay:
nptel.ac.in/courses/106101007

Relevant Topic for Self-study:

Large Language Model (LLM) Fine-Tuning: Fine-tune a pre-trained LLM for a specific task, such as a customer support agent or content generation



Society for Computer Technology & Research's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY

Multidisciplinary Minor Courses

Third Year B-Tech (TY B-Tech) AY (2026-27)
Electronics and Computer Engineering (E&CE)

[2605MD39]: Artificial Intelligence Lab (AIL)

Semester	Credits	Teaching Scheme	Examination Scheme
5	1	P: 2 Hrs./ Week	ESE (OR): 25 marks

Prerequisite: Students should have prior knowledge of

- Statistics, programming languages
- Machine Learning and algorithms

Course Objectives: The objective of this course is to provide students with

- Foundational understanding of concepts and techniques in Artificial Intelligence.
- Analyzing skills of various AI algorithms by comparing accuracy, efficiency, and applicability across domains.
- Applying NLP modeling techniques to develop real time application.
- Categorizing morphological and syntactic analysis for NLP models.

Course Outcomes: After completing this course, students will be able to

CO1: Explain fundamental principles of AI, intelligent agents, and basic search algorithms.

CO2: Design adversarial search strategies and constraint satisfaction solutions.

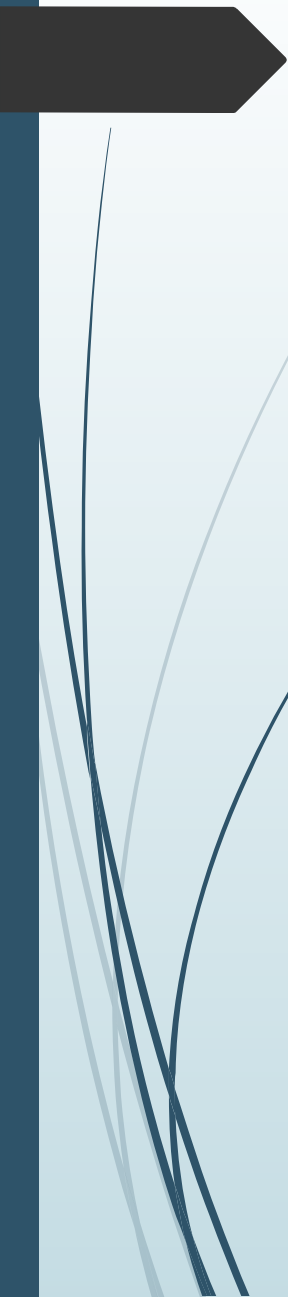
CO3: Perform morphological and syntactic analysis of NLP models.

CO4: Construct NLP modeling to develop real time application.

COURSE CONTENTS

Expt. No	Problem Statement	Hrs.	CO's
Group A (Any Four)			
1.	Implement A star (A*) Algorithm for any game search problem.	2	CO1
2.	Write a program for Water jug problem / Towers of Hanoi	2	CO1
3.	Write a program to solve n-queens problem using Hill-climbing / simulated annealing / A* algorithm	2	CO2
4.	Write a program for sorting using appropriate knowledge representation and reasoning techniques.	2	CO2
5.	Create and analyze a movie recommendation system using collaborative filtering.	2	CO2
Group B (Any Four)			
6.	Write a program for preprocessing Social Media Text and apply morphological analysis and subword tokenization in Indian Languages (Dataset: [Twitter / GujMORPH])	2	CO3
7.	Write a program to perform Text classification with HMM POS tagger using Viterbi decoding (Dataset: [CRAFT-SA])	2	CO3
8.	Write a program to build N-gram language model with contextual embeddings and fine tuning (Dataset: [IMDB Reviews Embeddings])	2	CO3
9.	Write a program to implement word and character embeddings using pre-	2	CO4

	trained GloVe embeddings (Dataset: [WSD Test])		
10.	Implement Named Entity Recognition in Healthcare to extract entities like diseases, symptoms, and drugs from clinical text. (Dataset: [Medical NER])	2	CO4
11.	Implement sentiment classification and extract topics from text data using extractive and abstractive text summarization (Dataset: [IMDB Reviews])	2	CO4
Group C			
12.	A Mini project: A group of 4 students can develop a small application in NLP with AI.	6	CO4
Text Books:			
T1.	S. J. Russell and P. Norvig, <i>Artificial Intelligence: A Modern Approach</i> , NJ, USA: Pearson, ISBN-13: 9780134610993, 4 th edition Hoboken		
T2.	J. Perkins, <i>Python 3 Text Processing with NLTK 3 Cookbook</i> , Birmingham, U.K.: Packt Publishing Ltd. ISBN-13: 9781782167853, 4 th edition		
Reference Books:			
R1.	S. Kaushik, <i>Artificial Intelligence</i> , India: Cengage Learning India, ISBN-13: 978-8131510995, 1 st edition		
R2.	D. Jurafsky and J. H. Martin, <i>Speech and Language Processing</i> , Pearson, ISBN-13: 978-9332518414, 3rd edition.		
R3.	Jacob Eisenstein, <i>Natural Language Processing</i> , MIT Press, ISBN-10: 0262042843, 1 st edition		
Relevant MOOCs Course (Course name and Weblink)			
1.	Artificial Intelligence search methods for problem solving by Prof. Deepak Khemani IIT Madras Link: https://onlinecourses.nptel.ac.in/noc21_cs79/preview		
2.	An Introduction to Artificial Intelligence by Prof. Mausam IIT Delhi An Introduction to Artificial Intelligence – Course, https://onlinecourses.nptel.ac.in/noc20_cs42/preview		
3.	Natural Language Processing by Prof. Pushpak Bhattacharyya, IIT Bombay: nptel.ac.in/courses/106101007		

A dark grey arrow points to the right from the left edge of the slide. Below it, several thin, curved lines in shades of blue and grey sweep upwards and to the right, creating a dynamic, abstract background element.

Module-I

Foundations of AI, Problem Solving and Search

Contents

□ Introduction to Artificial Intelligence

Definitions, foundations, history, state of the art, risks and benefits of AI

□ Intelligent Agents

Types, structure, environments, rationality, agent behavior, nature of environments

□ Problem-Solving

Problem characteristics, production systems, control strategies

□ Search Techniques

Uninformed search (breadth-first, depth-first), informed/heuristic search (hill climbing, best first), heuristic functions

□ Case Study

Application of AI in Healthcare Engineering

Introduction

Q. What is intelligence?

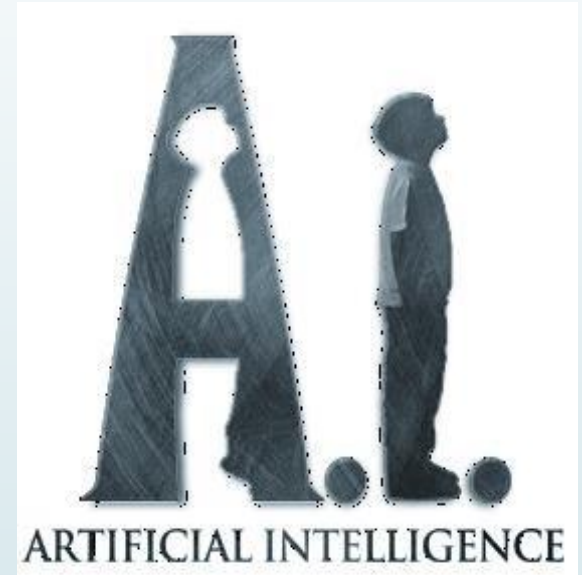
- “The capacity to learn and solve problems”
- “The computational part of the ability to achieve goals in the world.
- Varying kinds and degrees of intelligence occur in people, many animals and some machines.”
- Ability to think and act rationally.

What is Intelligence?

- Intelligence is a property of mind that includes many related mental abilities, such as the capabilities to
 - learn
 - think
 - reason
 - plan
 - solve problems

What is AI?

- **Artificial intelligence (AI)** is the intelligence exhibited by machines or software.
- It is an academic field of study which studies how to create computers and computer software that are capable of intelligent behavior.



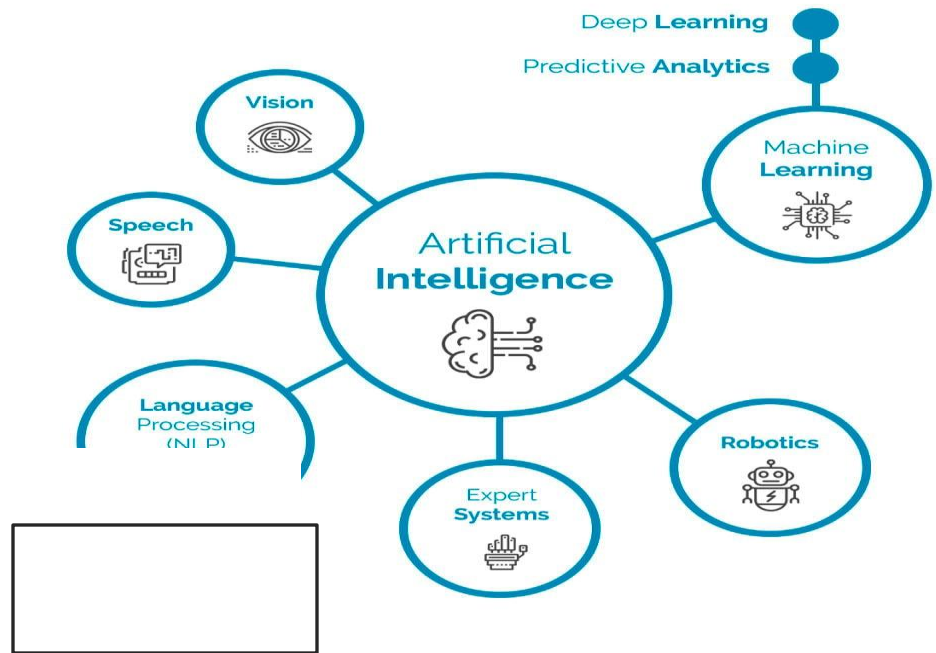
Definitions of AI



- A field that focuses on **developing techniques** to enable computer systems to perform activities that are considered intelligent (in humans and other animals).
[Dyer]
- The science and engineering of **making intelligent machines**, especially intelligent computer programs.
[McCarthy]
- Developing capabilities in machines such as
 - learn
 - think
 - reason
 - plan

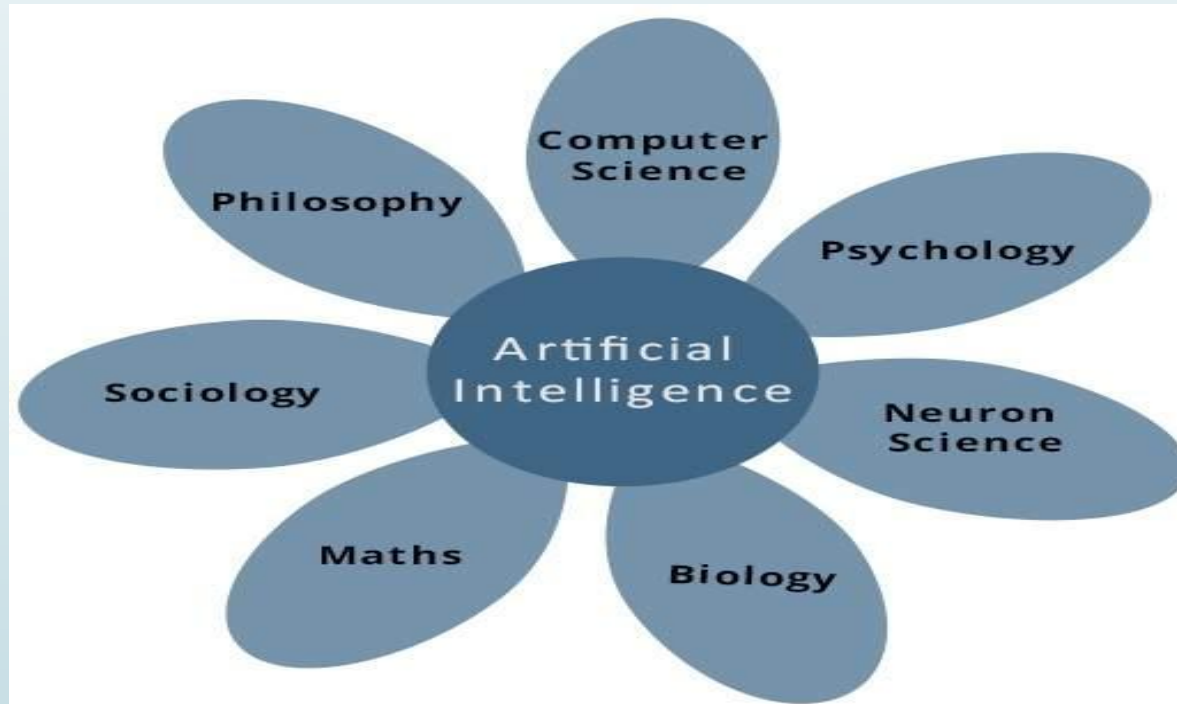
Introduction of AI

- **Artificial intelligence is the branch of computer science concerned with making computers behave like humans.**
- The term was coined in 1956 by John McCarthy.
- Artificial intelligence includes the following areas of specialization:
 - Games playing
 - Expert System
 - Natural Languages
 - Neural Network
 - Robotics



What Contributes to AI?

- AI is accomplished by **studying how human brain thinks, and how humans learn, decide, and work while trying to solve a problem**, and then using the outcomes of this study as a basis of developing intelligent software and systems.





Philosophy and Artificial Intelligence

- Philosophy forms the foundation of Artificial Intelligence because it explores the nature of **knowledge, reasoning, reality, and existence**. AI systems attempt to imitate human intelligence by using logical reasoning and decision-making principles derived from philosophy.
- **Philosophy** has influenced many core areas of **AI**, including:
 - Logical reasoning
 - Knowledge Representation
 - Expert Systems
 - Intelligent Agents
 - Automated Decision Making
 - Machine Ethics



Mathematics and Statistics in Artificial Intelligence

- Mathematics and Statistics provide the theoretical and computational foundation for Artificial Intelligence. They enable AI systems to represent knowledge, reason logically, learn from data, and make intelligent decisions under uncertainty.
- Formal Logic: Used to represent knowledge and perform logical reasoning. Algorithms: Mathematical principles are used to design efficient AI algorithms. Computation: Helps determine whether a problem can be solved computationally. Optimization: Finds the best solution among many alternatives. Linear Algebra: Represents data, vectors, matrices, and neural network computations. Calculus: Used in training machine learning and deep learning models through optimization techniques like gradient descent. Discrete Mathematics: Supports graph theory, search algorithms, and knowledge representation.
- Probability Theory: Handles uncertainty and incomplete information. Data Analysis: Identifies patterns and relationships in datasets. Inference: Draws conclusions from observed data. Prediction: Estimates future outcomes using statistical models. Machine Learning: Learns patterns from historical data to make predictions.

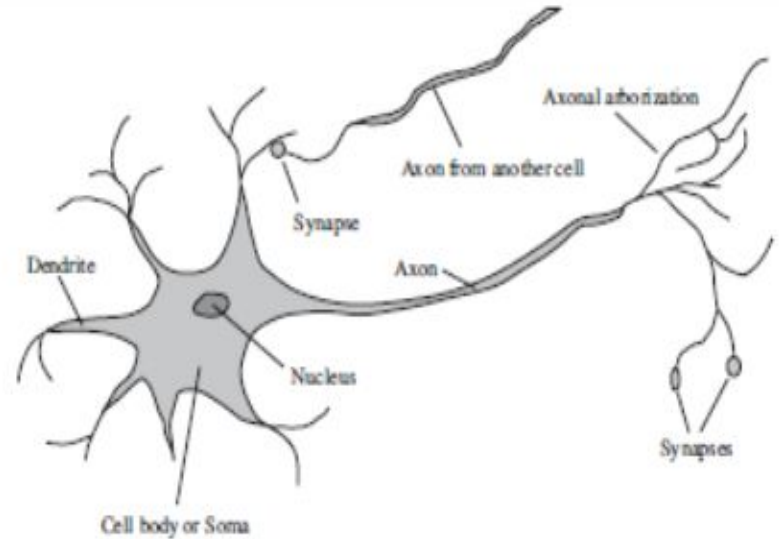


Neuroscience in Artificial Intelligence

- Neuroscience is the scientific study of the **nervous system**, especially the **human brain**. Since AI aims to build systems that exhibit intelligent behavior, understanding how the brain works provides valuable inspiration for designing intelligent algorithms.
- Helps understand how humans learn from experience. Explains how the brain processes sensory information. Provides insights into memory, reasoning, perception, and decision-making. Inspires the design of Artificial Neural Networks (ANNs) and Deep Learning models. Supports the development of intelligent systems capable of learning and adapting.

Biological Neuron

The parts of a nerve cell or neuron. Each neuron consists of a cell body, or soma, that contains a cell nucleus. Branching out from the cell body are a number of fibers called dendrites and a single long fiber called the axon. The axon stretches out for a long distance, much longer than the scale in this diagram indicates. Typically, an axon is 1 cm long (100 times the diameter of the cell body), but can reach up to 1 meter. A neuron makes connections with 10 to 100,000 other neurons at junctions called synapses. Signals are propagated from neuron to neuron by a complicated electrochemical reaction.



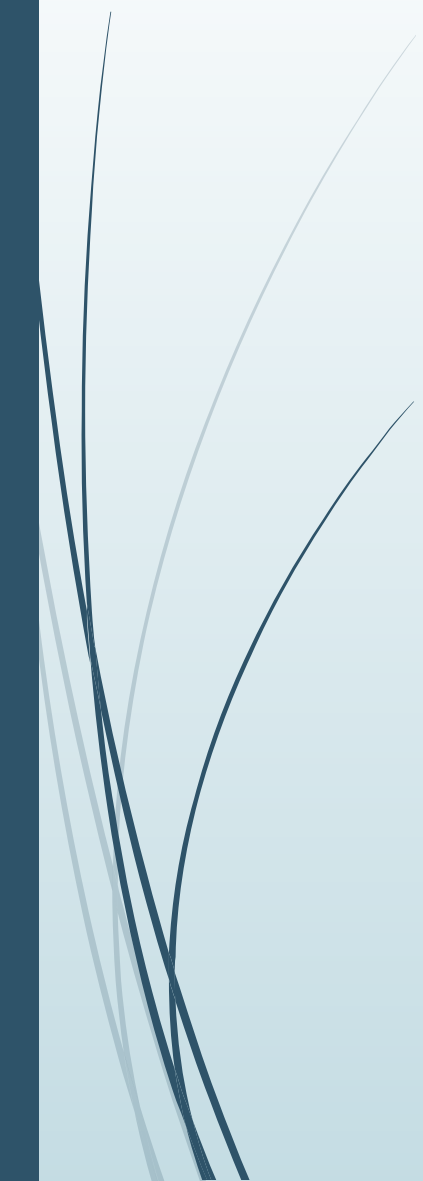
The signals control brain activity in the short term and also enable long-term changes in the connectivity of neurons. These mechanisms are thought to form the basis for learning in the brain. Most information processing goes on in the cerebral cortex, the outer layer of the brain. The basic organizational unit appears to be a column of tissue about 0.5 mm in diameter, containing about 20,000 neurons and extending the full depth of the cortex about 4 mm in humans).

Biological Neuron Vs Artificial Neuron

Biological Neuron	Artificial Neuron
Dendrites	Input Features
Synapse	Weights
Cell Body (Soma)	Weighted Sum Computation
Threshold	Activation Function
Axon	Output
Axonal Branches	Connections to Next Layer



Psychology and AI



- **Psychology** is the scientific study of **human behavior, thoughts, emotions, and mental processes**. **Cognitive Science** is an interdisciplinary field that studies how humans acquire, process, store, and use information.
- Artificial Intelligence draws inspiration from psychology because AI aims to develop systems that can **think, learn, reason, perceive, and make decisions** in ways similar to humans.



Computer Science & Engineering

- ❑ Computer Science and Engineering (CSE) forms the technological foundation of Artificial Intelligence. AI algorithms are designed, implemented, trained, and deployed using principles of computer science, supported by modern hardware and software technologies.
- ❑ CS provides:
 - ❑ Efficient algorithms
 - ❑ Programming languages
 - ❑ Data structures
 - ❑ Software engineering principles
 - ❑ Hardware support
 - ❑ Computing infrastructure

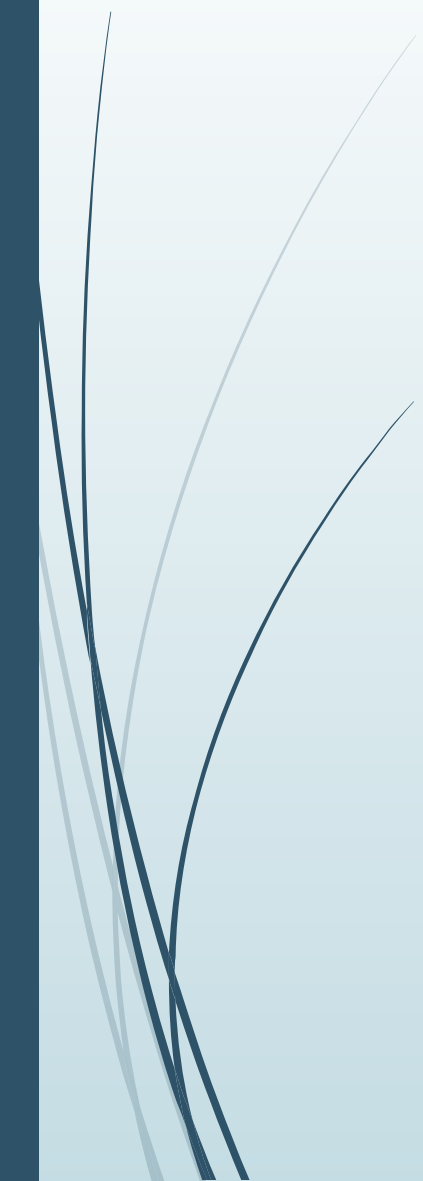


Control Theory and AI

- Control Theory is the branch of engineering and mathematics that studies how **dynamic systems** can be controlled automatically to achieve desired performance. It focuses on designing systems that can **monitor their behavior, compare it with the desired outcome, detect errors, and take corrective actions using feedback.**
- Feedback
- Error Detection
- Automatic Correction
- Stability
- Optimization
- Artificial Intelligence complements Control Theory by enabling systems to perceive their environment, learn from experience, make intelligent decisions, and adapt to changing conditions



Linguistics



- **Speech recognition** is a technology that enables a machine to understand spoken language and translate it into a machine-readable format.
- Speech Recognition
- Text-to-Speech (TTS)
- Natural Language Understanding (NLU)
- Natural Language Generation (NLG)
- Machine Translation
- Sentiment Analysis



ARTIFICIAL INTELLIGENCE



Philosophy: Where does knowledge come from?



Linguistics: How does language relate to thought?



Neuroscience: How do our brains process information?



Behavioral Economics: How do you make decisions to maximize utility?

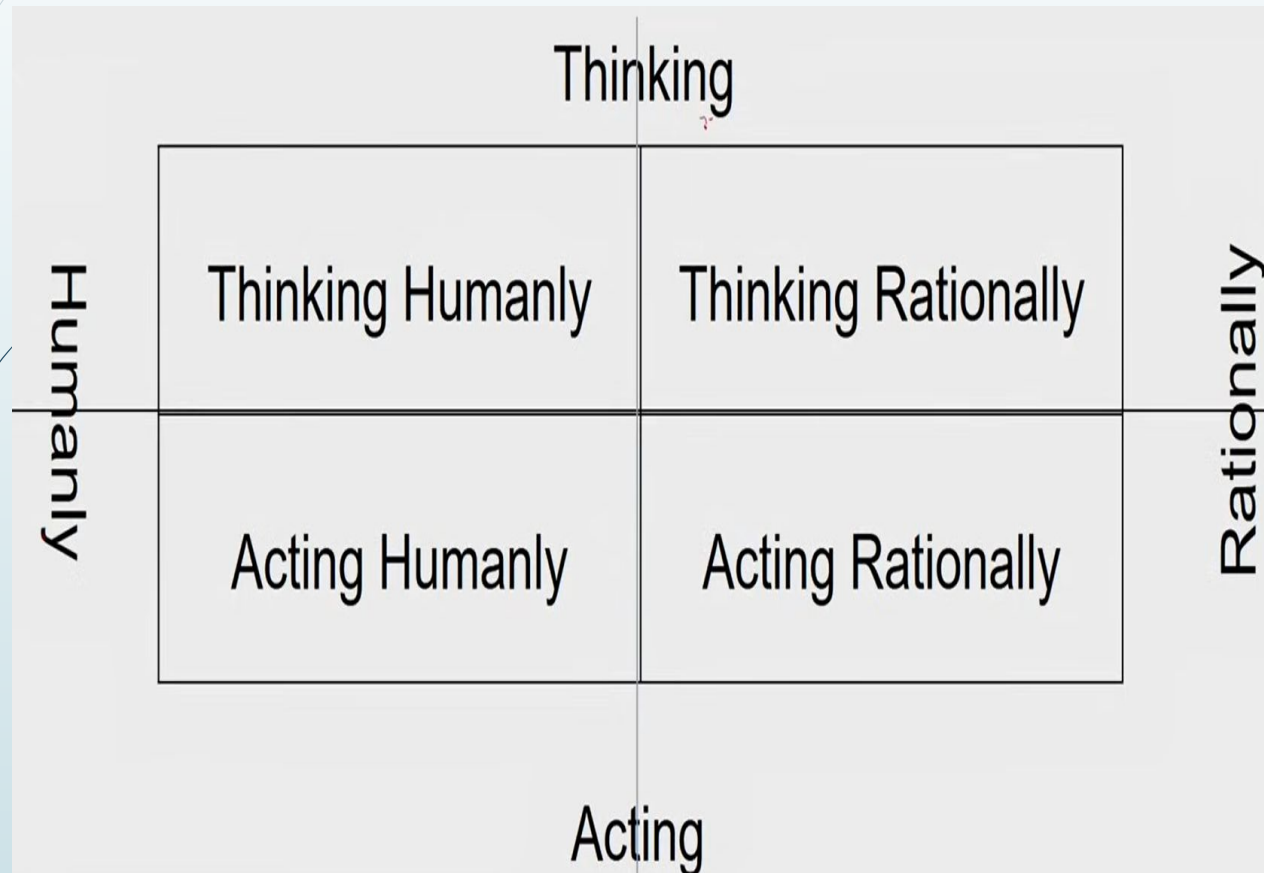


Mathematics: What can be computed?



Computer Science: How can we build an efficient computer?

Dimensions of AI



introduced by **Stuart Russell and Peter Norvig** in *Artificial Intelligence: A Modern Approach (AIMA)*

1. Think like Human

- model human cognition

1960s "cognitive revolution": information-processing
psychology

Requires **scientific theories** of internal activities
of the **brain**

General Problem Solver

3. Act Rationally

- doing the right thing

Rational behavior: **doing the right thing** - one which is
expected to maximize goal achievement, given the available
information

Solve Problem

2. Think Rationally

- formalize the inference process.

Several **Greek schools** developed various forms of
logic: notation and *rules of derivation for*
thoughts; may or may not have proceeded to the idea of
mechanization

Direct line through mathematics and philosophy to modern AI

Logical Thinking

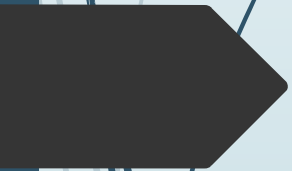
4. Act like Human

- exhibit human behaviour

Creating **machines that perform functions**
that require intelligence when the **same**
tasks are performed by people.

Done like Human

History of AI



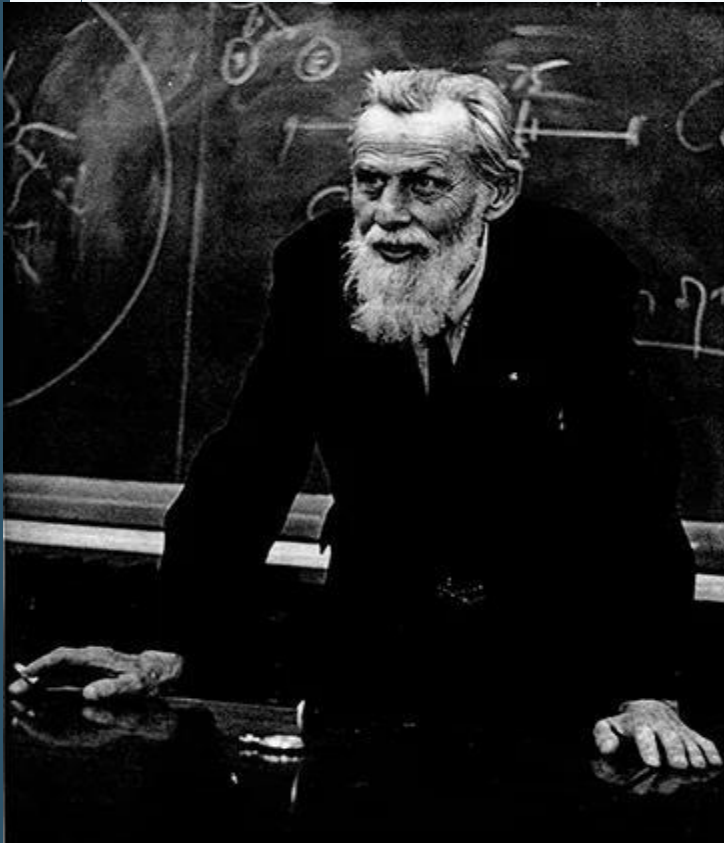
Aristotle	the first to formulate a precise set of laws governing the rational part of the mind and allowed one to generate conclusions mechanically, given initial premises.
Ramon Lull	the idea that useful reasoning could actually be carried out by a mechanical artifact.
Thomas Hobbes (1588–1679)	proposed that reasoning was like numerical computation, that “we add and subtract in our silent thoughts.”
Wilhelm Schickard (1592–1635)	The first known calculating machine was constructed around 1623 by the German scientist
René Descartes (1596–1650)	gave the first clear discussion of the distinction between mind and matter and of the problems that arise
Hume, 1739	proposed what is now known as the principle of INDUCTION induction: that general rules are acquired by exposure to repeated associations between their elements.
Rudolf Carnap (1891–1970),	developed the LOGICAL POSITIVISM doctrine of logical positivism. This doctrine holds that all knowledge can be characterized by OBSERVATION logical theories connected,

Carnap and
Carl Hempel
(1905–1997)

The confirmation theory attempted to analyze the acquisition of knowledge from experience.

*The final element in the philosophical picture
of the mind is the connection between
knowledge and action*

Birth of AI (1943-56)

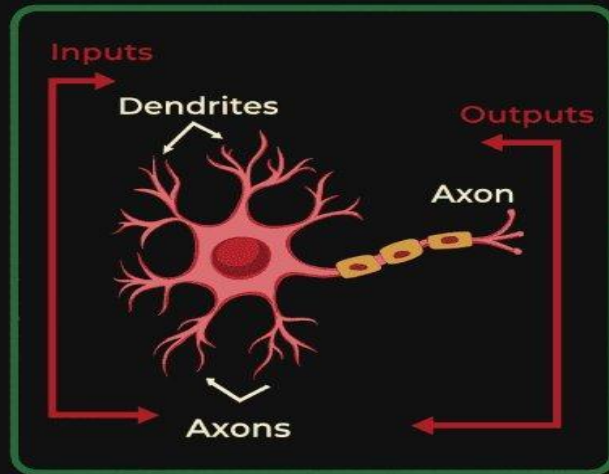


- Warren McCulloch & Walter Pitts (1943): ANN with on-off neurons
 - Neurons triggered by sufficient neighbors
 - Showed that any computable function computable with some network
 - Logical connectives implementable

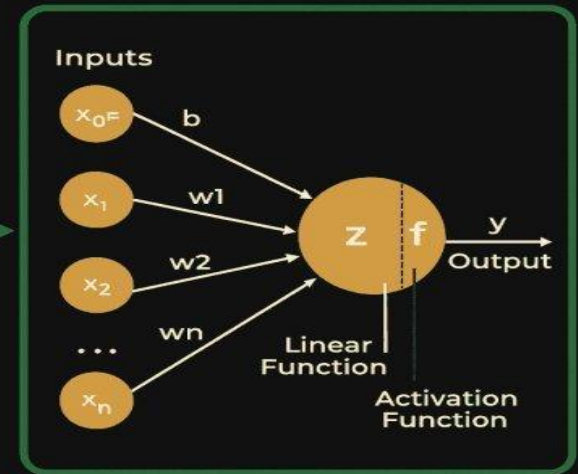
journey began more than 80 years ago with a very simple question—How does the human brain work?

"Scientists tried to copy the working of the human brain using mathematics and electronics. This gave birth to Artificial Intelligence.

BIOLOGICAL NEURON



ARTIFICIAL NEURON



- What is the smallest functional unit of the brain-Neuron
- Can we create a mathematical neuron?
- first Artificial Neural Network (ANN) model



History of AI



- 1943:** McCulloch & Pitts created the first mathematical model of an artificial neuron.
- 1949:** Donald Hebb explained how learning happens by strengthening neural connections.
- 1950:** Alan Turing and Claude Shannon showed that computers could potentially solve intelligent tasks like chess.
- 1951:** SNARC became the first neural-network-based learning machine.
- 1956:** The Dartmouth Conference officially introduced the term **Artificial Intelligence**.

Alan Turing and Claude Shannon

- Can Machine Think?
- **1950:** Alan Turing and Claude Shannon showed that computers could potentially solve intelligent tasks like chess.

Birth of AI...

- Dartmouth 1956 workshop for 2 months
 - Term “artificial intelligence”
 - Fathers of the field introduced
- *Logic Theorist*: program for proving theorems by Alan Newell & Herbert Simon

Thomas Hobbes: The Grandfather of AI

It was the English philosopher Thomas Hobbes (1588-1679) who first put forward the view that **thinking is the manipulation of symbols**.

Galileo had said that all reality is mathematical in the sense that everything is made up of particles, and our sensing of smell or taste was how we reacted to those particles.

Hobbes extended this notion to say that thought too was made up of (expressed in) particles which the thinker manipulated.

Hobbes was influenced by Galileo. Just as geometry could represent motion, thinking could be done by manipulation of mental symbols.



How AI got its name

The name *artificial intelligence* is credited to John McCarthy who, along with Marvin Minsky and Claude Shannon (1916–2001), organized the *Dartmouth Conference* in 1956.

Early enthusiasm (1952-69)

- Claims: computers can do X
- *General Problem Solver*, Newell & Simon
 - Intentionally solved puzzles in a similar way as humans do (order of subgoals, etc)
- *Geometry Theorem Prover*, Herbert Gelernter, 1959
- Arthur Samuel's learning checkers program 1952
- LISP, time sharing, Advice taker: McCarthy 1958
- Integration, IQ geometry problems, algebra stories
- Blocks world: vision, learning, NLP, planning
- Adalines [Widrow & Hoff 1960], perceptron convergence theorem [Rosenblatt 1962]

A dose of reality (1966-74)

- Simple syntactic manipulation did not scale
 - ELIZA (example rule: if sentence contains “mother”, then say: “tell me more about your family”)
 - However, sometimes such bots (e.g. Julia) can fool humans
 - “the spirit is willing but the flesh is weak” -> “the vodka is good but the meat is rotten”
- Intractability
- Machine evolution did not scale
- *Perceptrons* book with negative result on representation capability of 1-layer ANNs [Minsky & Papert]

ELIZA [Weizenbaum 1966]

Men are all alike.

IN WHAT WAY

They're always bugging us about some other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time

I AM SORRY TO HEAR THAT YOU ARE DEPRESSED

It's true that I am unhappy.

DO YOU THINK COMING HERE WILL HELP YOU NOT BE UNHAPPY

I need some help, that much seems certain

WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP

Perhaps I could learn to get along with my mother.

TELL ME MORE ABOUT YOUR FAMILY

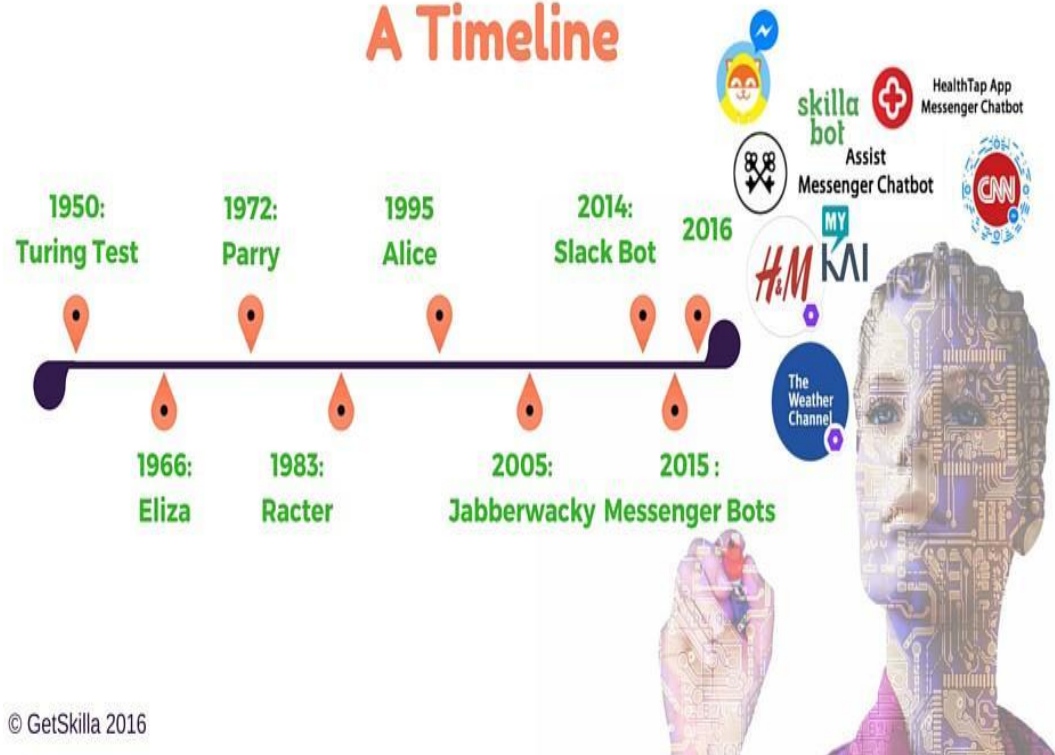
My mother takes care of me.

WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU

Chatbot

† <https://web.njit.edu/~ronkowitz/eliza.html>

History of Chatbots: A Timeline



ELIZA...

HOW DO YOU DO. PLEASE STATE YOUR PROBLEM

I'm wondering whether computers can think.

WHY DO YOU MENTION COMPUTERS

Since you are one I thought I would ask you - can you think?

DID YOU THINK THEY MIGHT NOT BE ONE YOU THOUGHT YOU WOULD ASK I -CAN I THINK

Let's talk just about you - not me. Can you think?

YOU LIKE TO THINK I - NOT YOU - DON'T YOU

Pardon?

WHAT DOES THAT SUGGEST TO YOU?

What does it suggest to you?

TELL ME ABOUT WHAT DOES IT SUGGEST TO I

That's nonsense - which of us do you mean by I?

WHY DO YOU THINK I MEAN BY YOU

The question meant to which of us is the word referring.

SUPPOSE THE QUESTION MEANT TO WHICH OF US WERE NOT THE WORD REFERRING

I think you are an extremely stupid program.

PERHAPS YOU WOULD LIKE TO BE AN EXTREMELY STUPID PROGRAM

Knowledge-based systems (1969-79)

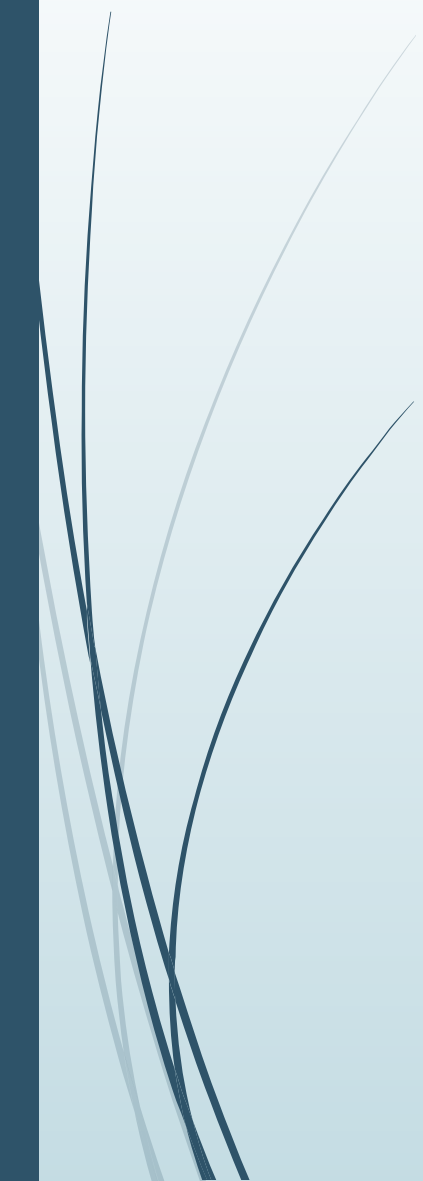
- *DENDRAL*: molecule structure identification [Feigenbaum et al.]
 - Knowledge intensive
- *Mycin*: medical diagnosis [Feigenbaum, Buchanan, Shortliffe]
 - 450 rules; knowledge from experts; no domain theory
 - Better than junior doctors
 - Certainty factors
- *PROSPECTOR*: drilling site choice [Duda et al]
- Domain knowledge in NLP
- Knowledge representation: logic, frames...

AI becomes an industry (1980-88)

- *R1*: first successful commercial expert system, configured computer systems at DEC; saved 40M\$/year
- 1988: DEC had 40 expert systems, DuPont 100...
- 1981: Japan's 5th generation project
- Software tools for expert systems: Carnegie Group, Inference, Intellicorp, Teknowledge
- LISP-specific hardware: LISP Machines Inc, TI, Symbolics, Xerox
- Industry: few M\$ in 1980 -> 2B\$ in 1988



Return of ANNs (1986-)

- Mid-1980s, different research groups reinvented backpropagation (originally from 1969)
 - Disillusionment on expert systems
 - Fear of AI winter
- 

Recent events (1987-)

- Rigorous theorems and experimental work rather than intuition
- Real-world applications rather than toy domains
- Building on existing work
 - E.g. speech recognition
 - Ad hoc, fragile methods in 1970s
 - Hidden Markov models now
 - E.g. planning (unified framework helped progress)
- Normative system design
 - Belief networks & probabilistic reasoning
 - Reinforcement learning
 - Multiagent systems
 - Resource-bounded reasoning

The emergence of intelligent agents (1995–present)

1

- One of the most important environments for intelligent agents is the Internet.

2

- AI systems have become so common in Web-based applications that the “-bot” suffix has entered everyday language

3

- applications that are good at a specific task, such as driving a car, playing chess, or recognizing speech.



The availability of very large data sets (2001–present)

- *the emphasis has been on the algorithm as the main subject of study.*
- *word-sense disambiguation: given the use of the word “plant” in a sentence, does that refer to flora or factory?*
- *human-labeled examples combined with machine learning algorithms.*

History of AI

□ 1943: early beginnings

- McCulloch & Pitts: Boolean circuit model of brain

□ 1950s: initial promise

- Turing's "Computing Machinery and Intelligence"
- Samuel's checkers program
 - First program of Machine Learning
- Early AI programs

□ 1955-65: Great enthusiasm!

- Dartmouth meeting
 - "Artificial Intelligence" name adopted
- Newell and Simon: GPS, general problem solver
- Gelertner: Geometry Theorem Prover
- McCarthy: invention of LISP

□ 1969—85: Adding domain knowledge

- Development of knowledge-based systems
- Success of rule-based expert systems,
 - E.g., DENDRAL, MYCIN
 - But were brittle and did not scale well in practice

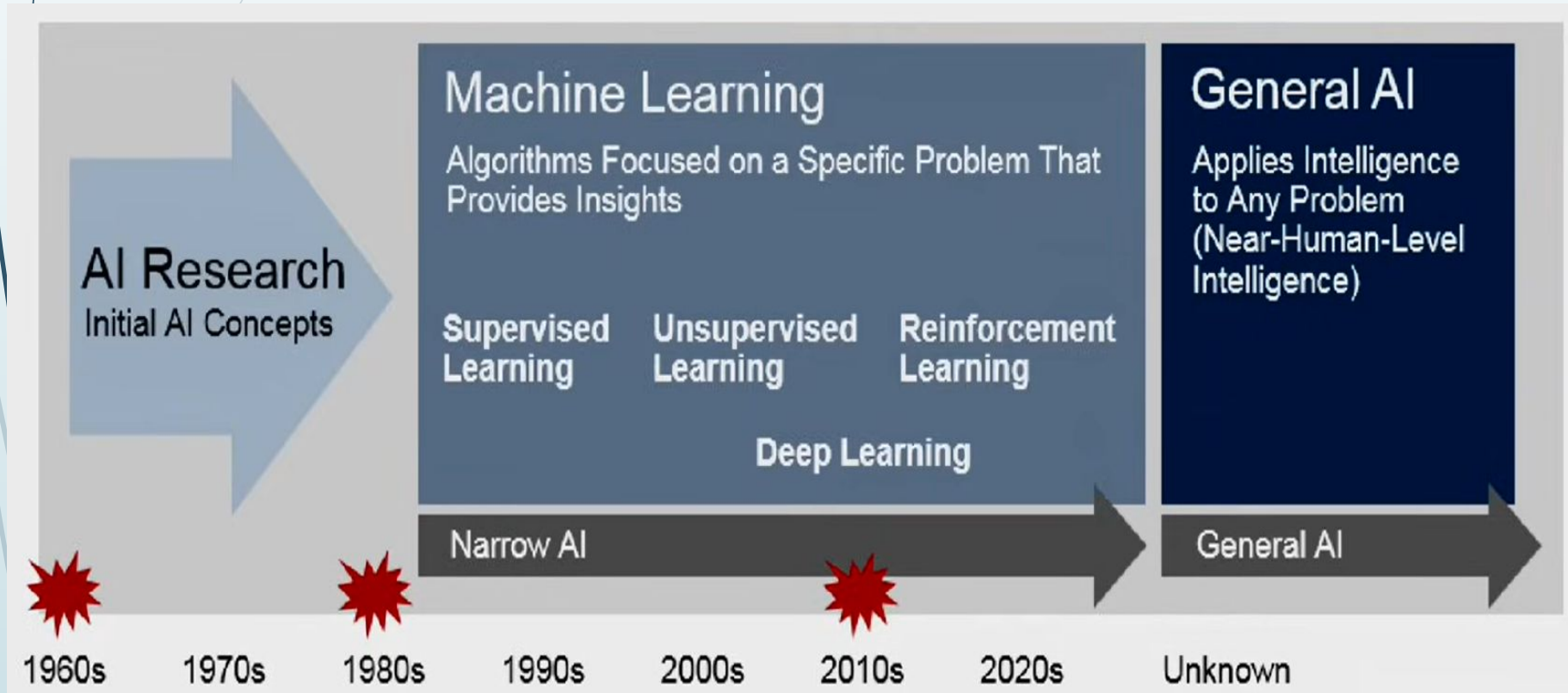
□ Beginning 1990: Role of uncertainty

- Bayesian networks
 - knowledge representation framework

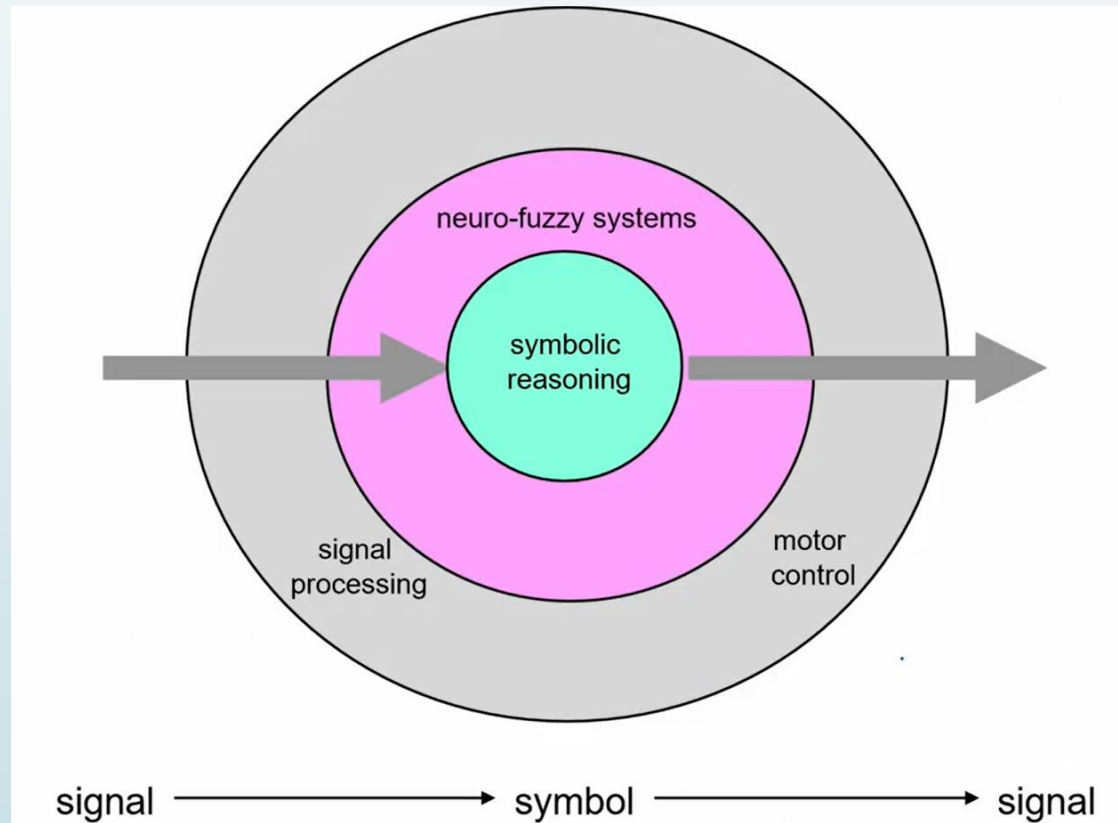
□ Beginning 1995: AI as Science

- Integration of learning, reasoning, knowledge representation

History of AI



Information Processing in View of AI



Some definitions of artificial intelligence

Thinking Humanly

"The exciting new effort to make computers

think . . . machines with minds, in the

full and literal sense."

(Haugeland, 1985)

"[The automation of] activities that we associate with human thinking, activities

such as decision-making, problem solving,

learning . . ." (Bellman, 1978)

Thinking Rationally

"The study of mental faculties through the

use of computational models."

(Charniak and McDermott, 1985)

The study of the computations that make

it possible to perceive, reason, and act."

(Winston, 1992)

Acting Humanly

"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)

Acting Rationally

"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)



THE STATE OF THE ART

□ What can AI do today?

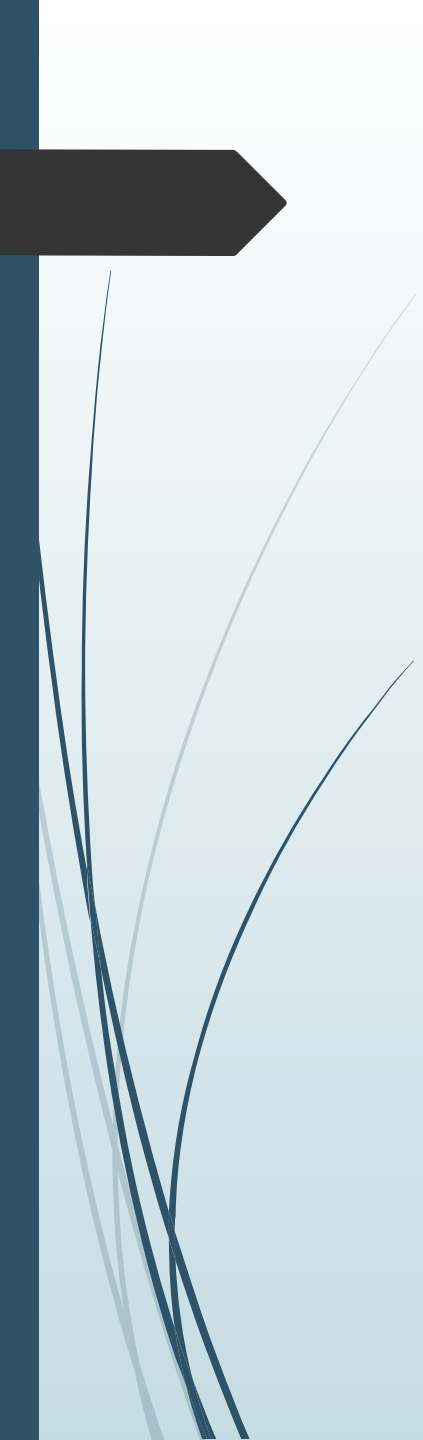
1. **Robotic vehicles:** A driverless robotic car named STANLEY.
2. **Speech recognition:** A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

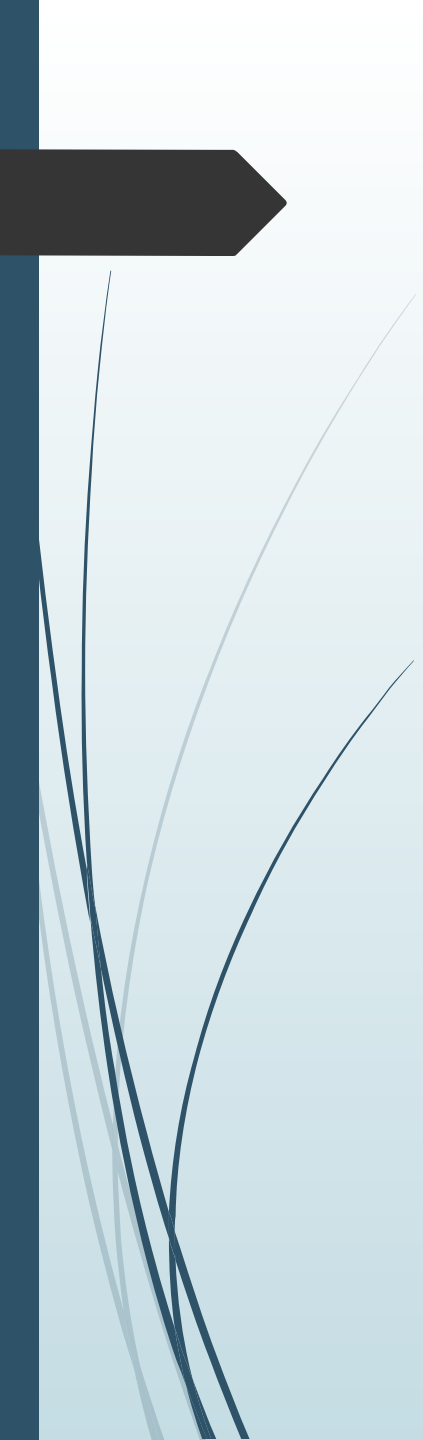


3. Autonomous planning and scheduling: A hundred million miles from Earth, NASA's

Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft

4. Game playing: IBM's DEEP BLUE became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match

- 
- 5. Spam fighting:** Each day, learning algorithms classify over a billion messages as spam, saving the recipient from having to waste time deleting what, for many users, could comprise 80% or 90% of all messages.
 - 6. Logistics planning:** During the Persian Gulf crisis of 1991, U.S. forces deployed a Dynamic Analysis and Replanning Tool, DART (Cross and Walker, 1994), to do automated logistics planning and scheduling for transportation.

- 
7. **Robotics:** The iRobot Corporation has sold over two million Roomba robotic vacuum cleaners for home use.
 8. **Machine Translation:** A computer program automatically translates from Arabic to English,

AI Applications

- Autonomous Planning & Scheduling:
 - Autonomous rovers.



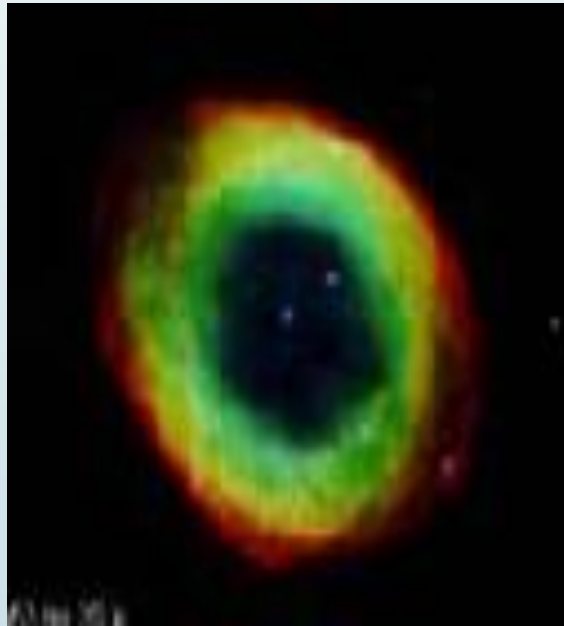
AI Applications

- Autonomous Planning & Scheduling:
 - Telescope scheduling



AI Applications

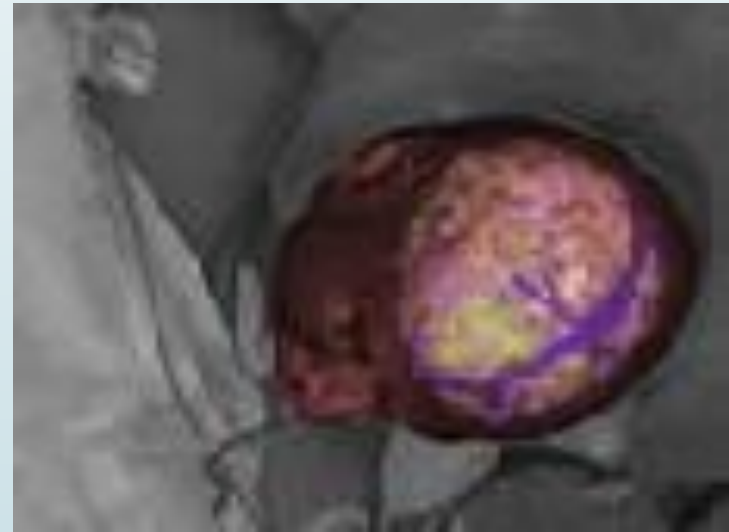
- Autonomous Planning & Scheduling:
 - Analysis of data:



AI Applications

□ **Medicine:**

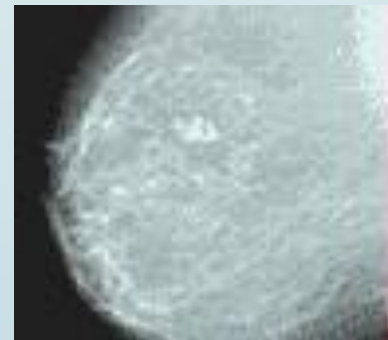
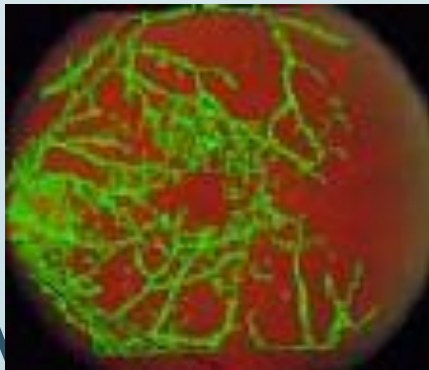
- Image guided surgery



AI Applications

□ **Medicine:**

- Image analysis and enhancement



AI Applications

□ **Transportation:**

○ **Autonomous vehicle control:**



AI Applications

- **Transportation:**
 - **Pedestrian detection:**



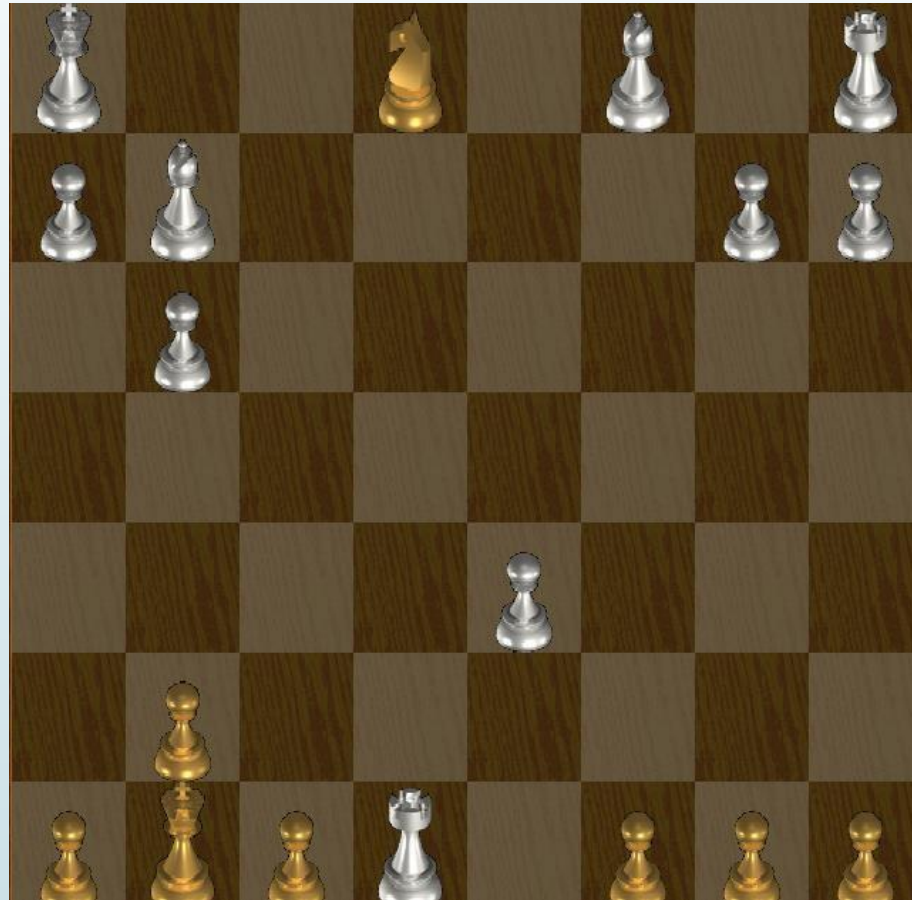
AI Applications

Games:



AI Applications

 Games:



AI Applications

□ Robotic toys:



Benefits of AI

1. Increased Efficiency

- ☐ Automates repetitive tasks, Works continuously without fatigue, Improves productivity in industries such as manufacturing, logistics, and customer service.

2. Better Decision-Making

- ☐ Analyzes large amounts of data quickly , Identifies patterns humans may miss, Supports decision-making in healthcare, finance, and business.

3. Advances in Healthcare

- ☐ Assists in disease diagnosis, Accelerates drug discovery, Enables personalized treatment plans.

4. Enhanced Education

- ☐ Provides personalized learning experiences, Offers tutoring and learning support, Makes education more accessible worldwide.

5. Scientific and Technological Progress

- ☐ Helps researchers process complex data, Contributes to breakthroughs in fields such as Genetics, climate science, and space exploration.

6. Improved Accessibility

- ☐ Supports people with disabilities through speech recognition, translation, and assistive technologies, Enables real-time communication across

Risks and Benefits of AI

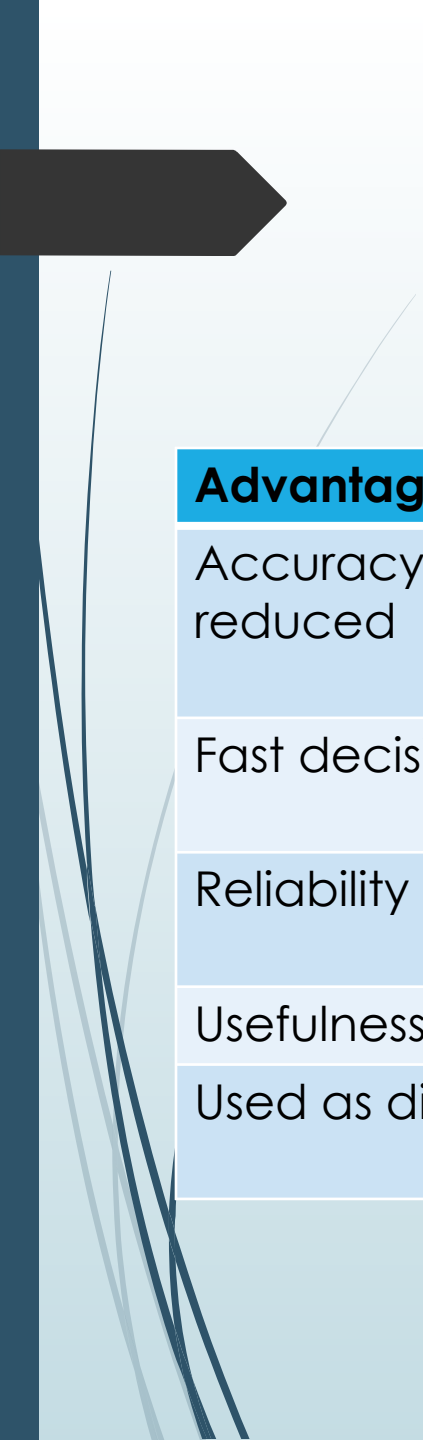
❖ *AI machines use machine learning algorithms to mimic the cognitive abilities of human beings and solve a simple or complex problem.*

❑ **Benefits of Artificial Intelligence:**

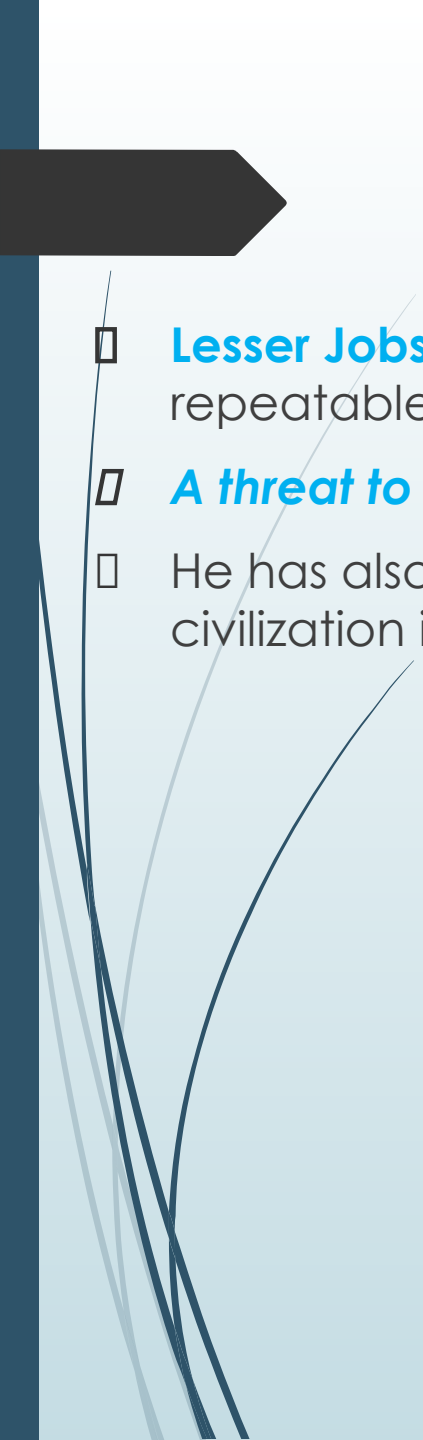
- ❑ Increase work efficiency. ...
- ❑ Work with high accuracy. ...
- ❑ Reduce cost of training and operation. ...
- ❑ Improve Processes. ...

Risks of Artificial Intelligence

- ❑ Risks of Artificial Intelligence. ...
- ❑ AI is Unsustainable. ...
- ❑ Lesser Jobs.
- ❑ **AI is Unsustainable**
- ❑ Intelligent machines have characteristically high computing powers contributed by an array of several processors.
- ❑ These computer chips have rare earth materials like **Selenium** as a major constituent. Besides, the batteries of such machines run on **Lithium**, again a rare element in earth's crust.
- ❑ The increased mining of these materials is irreversibly **damaging our environment** at a rapid pace.



Advantages of AI	Disadvantages of AI
Accuracy is higher and error is reduced	Cost increases as demand of high computational machines increases
Fast decision making	Can't think beyond limit(e.g. chess)
Reliability is more	No feeling in machine will perform dedicated tasks only
Usefulness in risky areas	More dependency on machines
Used as digital assistant	No original thinking available depends on feed thinking

- 
- ❑ **Lesser Jobs:** There is no doubt that machines do routine and repeatable tasks much better than humans.
 - ❑ ***A threat to Humanity(?):***
 - ❑ He has also stated publicly that AI is the biggest threat to human civilization in the future.

□ Risks of AI

□ 1. Job Displacement

Automation may replace certain jobs.

Workers may need retraining and new skills.

Some industries could experience significant workforce disruption.

2. Bias and Discrimination

AI systems can inherit biases from training data.

May produce unfair outcomes in hiring, lending, or law enforcement.

3. Privacy Concerns

Large-scale data collection can threaten personal privacy.

Misuse of personal information may occur without proper safeguards.

4. Misinformation and Deepfakes

AI can generate convincing fake images, videos, and text.

May be used to manipulate public opinion or spread false information.

5. Security Risks

Cybercriminals can use AI to automate attacks.

AI systems themselves may be vulnerable to hacking or manipulation.

6. Lack of Transparency

Some AI models operate as "black boxes," making decisions difficult to explain.

This can reduce trust and accountability.

7. Long-Term Safety Concerns

Highly capable AI systems could be misused for harmful purposes.

Ensuring human oversight and alignment with human values remains an ongoing challenge.



Conclusion



- **Every coin has two sides and Artificial Intelligence is no different. The rise of AI-powered machines has undoubtedly eased our lives in many applications even today.**
- **"The future of AI is not about replacing humans; it is about humans and AI working together to solve problems more efficiently, ethically, and intelligently."**
- **"AI is neither inherently good nor bad—it is a powerful tool. Its impact depends on how responsibly humans design, develop, and use it."**

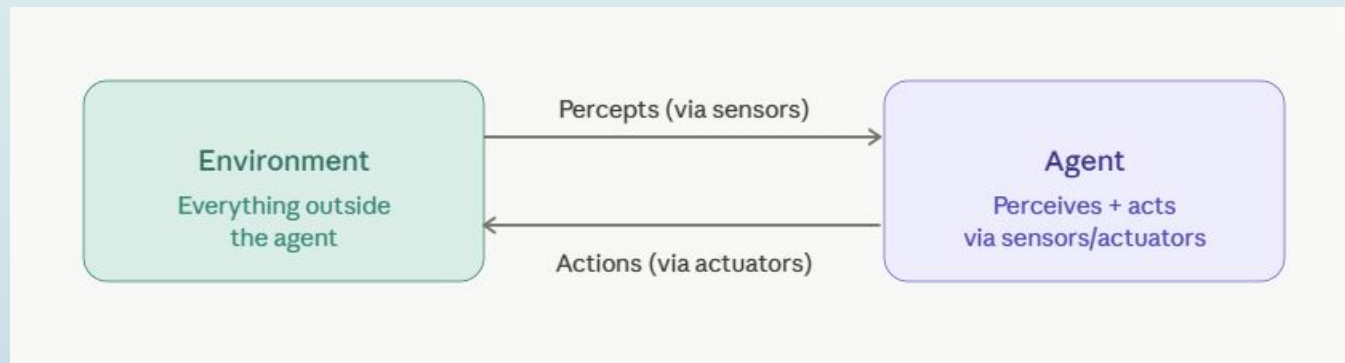


Intelligent Agents

- to develop a small set of design principles for building successful agents—systems that can reasonably be called **intelligent**.
- An **agent is anything that can be viewed as perceiving its environment through sensors and** acting upon that environment through **actuators**.
- **Agents:** smart systems that can sense, think, and act just like us
- **E.g.** Google Maps, Alexa, or seen a self-driving car video,
- humans

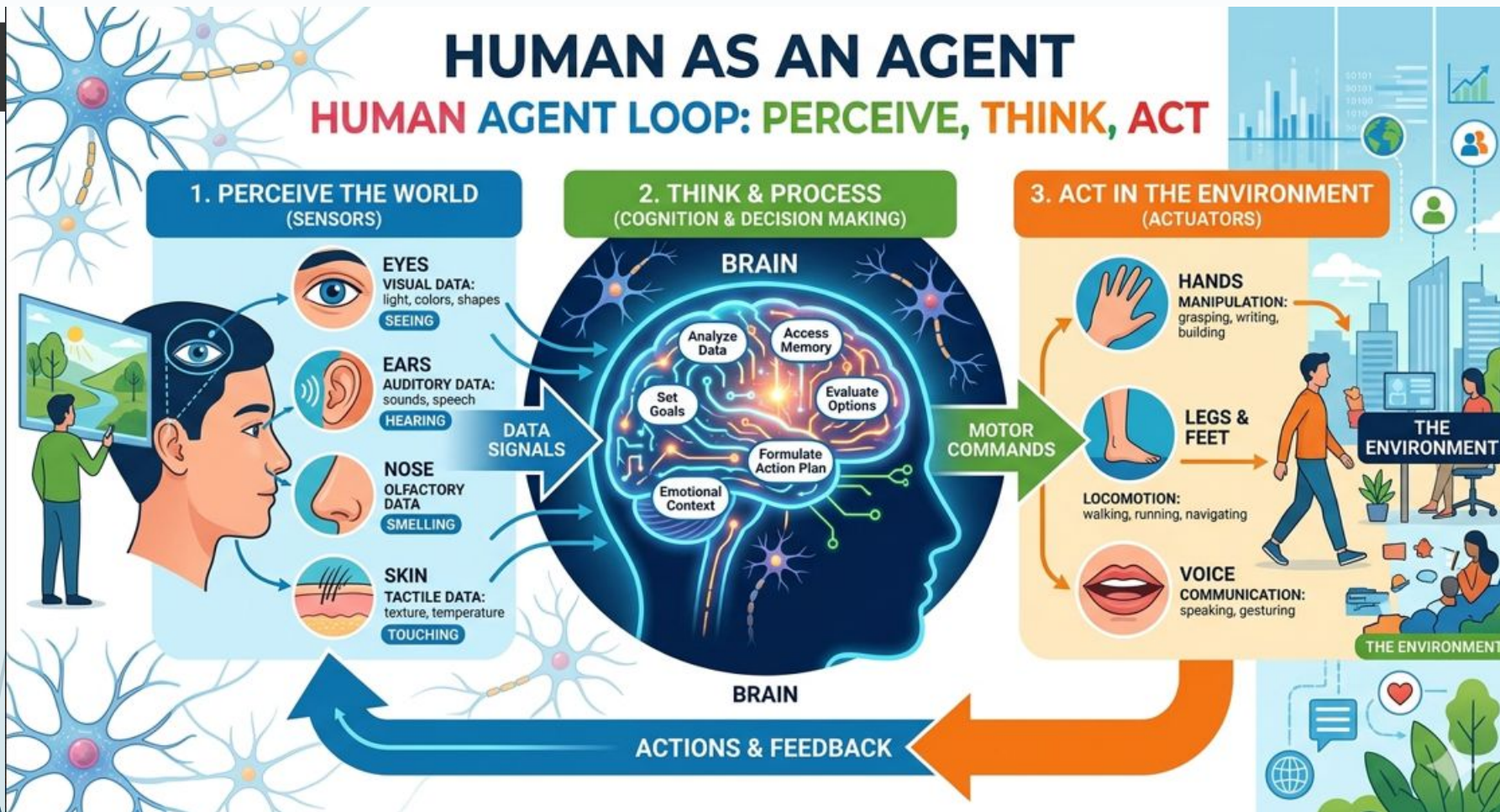
What Is an Intelligent Agent?

- An agent is anything that perceives its environment through sensors and acts upon it through actuators. The word "intelligent" adds the expectation that it does so *rationally*.
- Percept → Agent Function → Action



HUMAN AS AN AGENT

HUMAN AGENT LOOP: PERCEIVE, THINK, ACT



AI agents work the same way — they sense, decide, and act!

Intelligent Agents

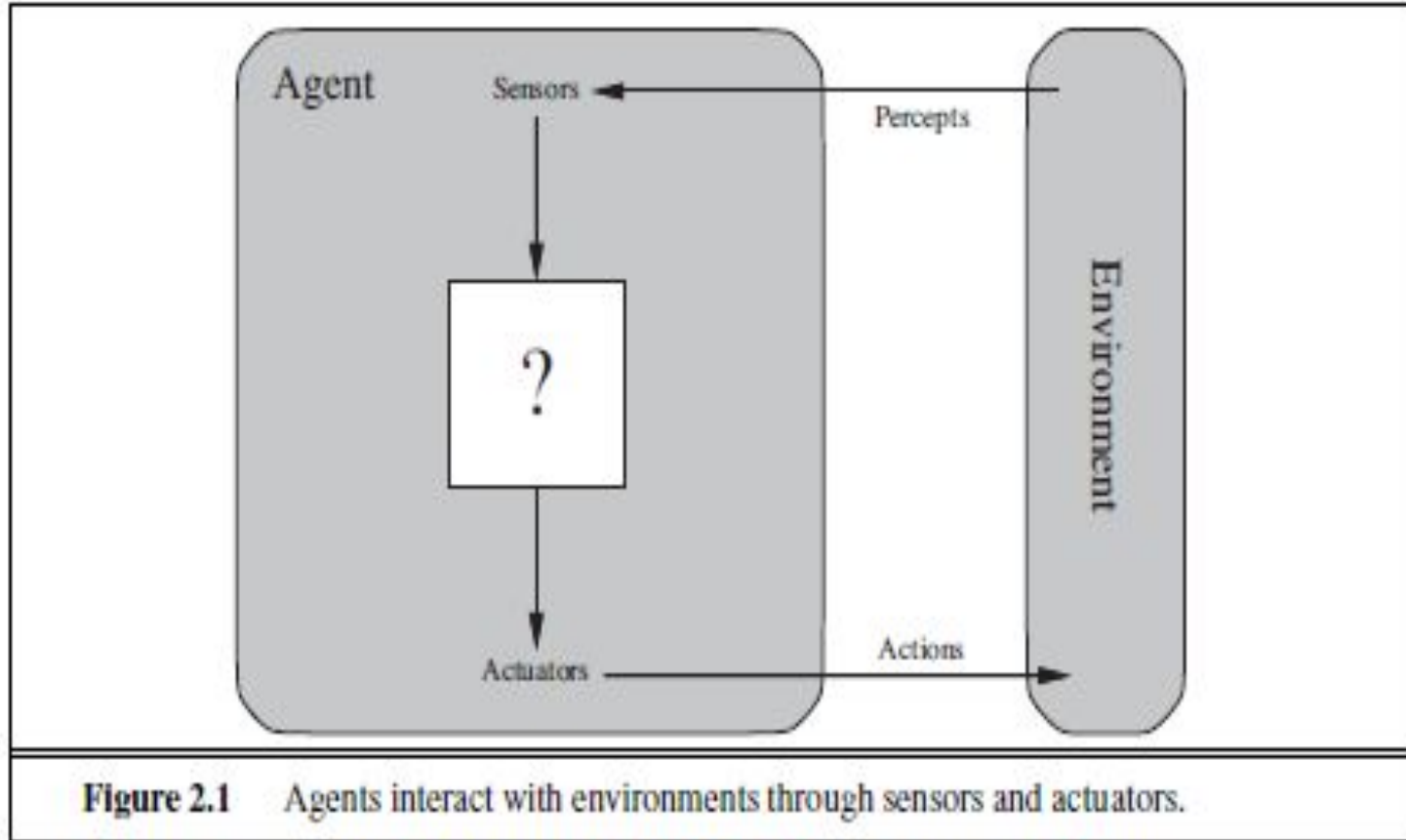
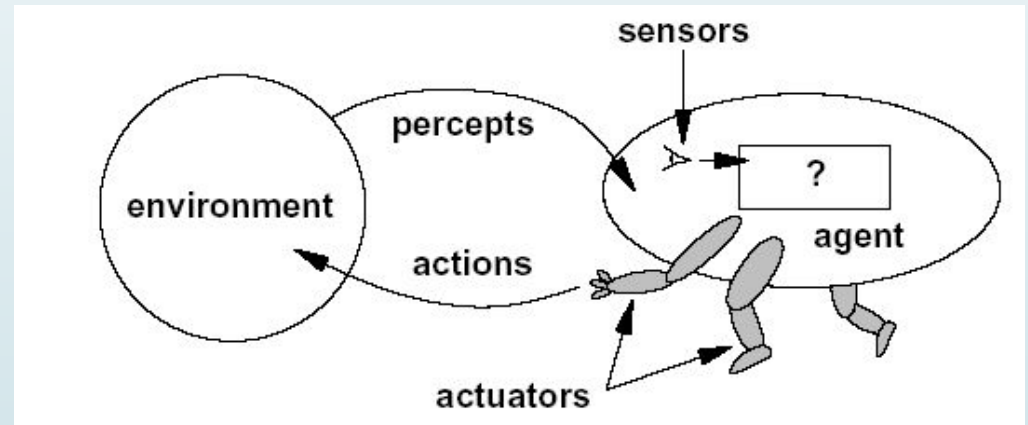


Figure 1: Agent

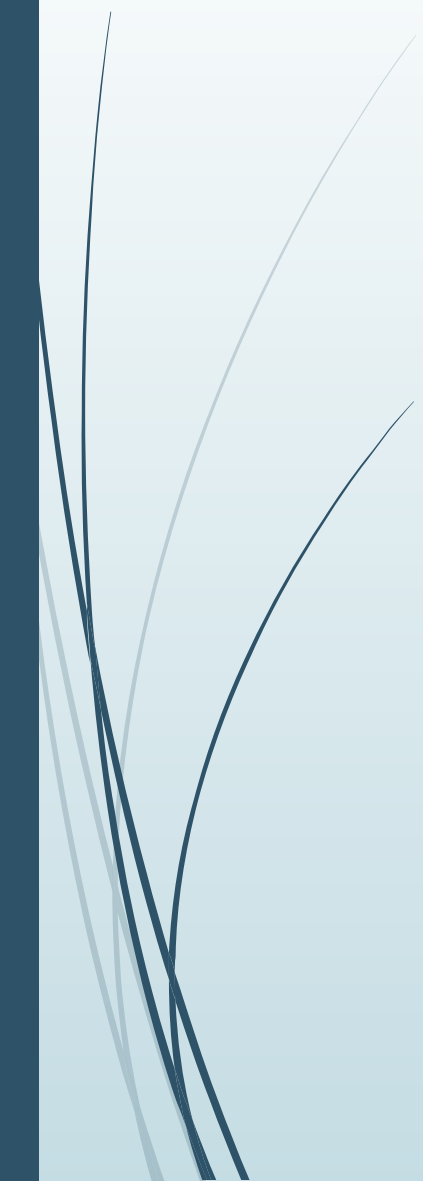
Example 1

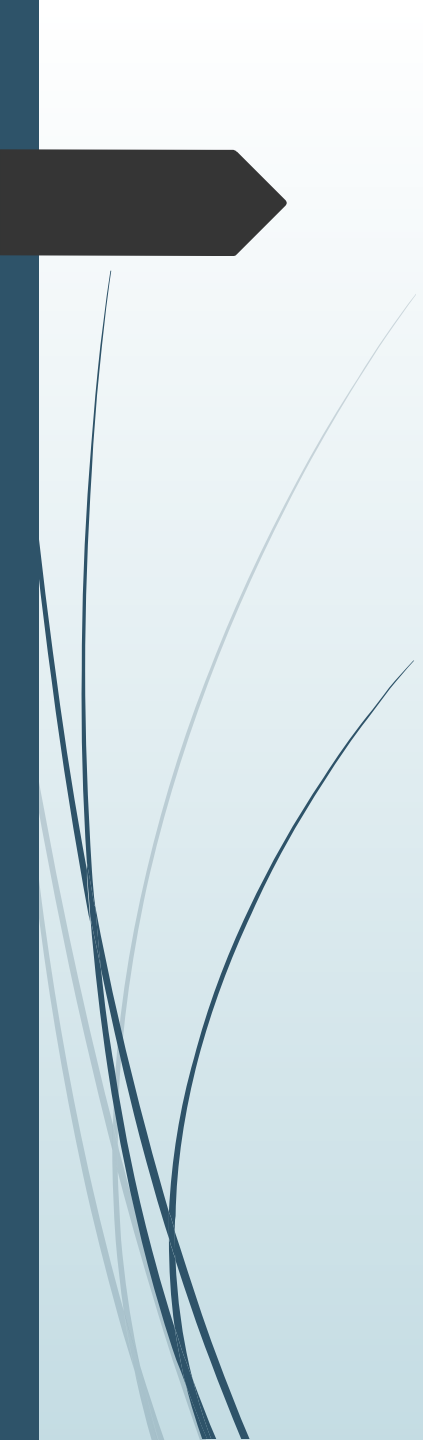
- A human agent has :
 - Sensors: eyes, ears, and other organs.
 - Actuator: hands, legs, mouth, and other body part.
 - A robotic agent might have:
 - Sensors: Cameras,...
 - Actuator: motors
 - A Software Agent
 - Sensors:
 - Actuator:
 - The agent function maps from percepts histories to actions
- $F: P \rightarrow A$





How Agent works?

- four main components —
 - **Percept**
 - **Action**
 - **Goal**
 - **Performance Measure.**
- 



1. **Percept** — What the Agent Sees

The **percept** is the *input* that the agent receives from its environment through sensors.

2. **Action** — What the Agent Does

The **action** is what the agent performs using its actuators (motors or effectors).

Based on its perception, the agent decides what to do next.

3. **Goal** — What the Agent Wants to Achieve

Every intelligent agent works toward a goal.

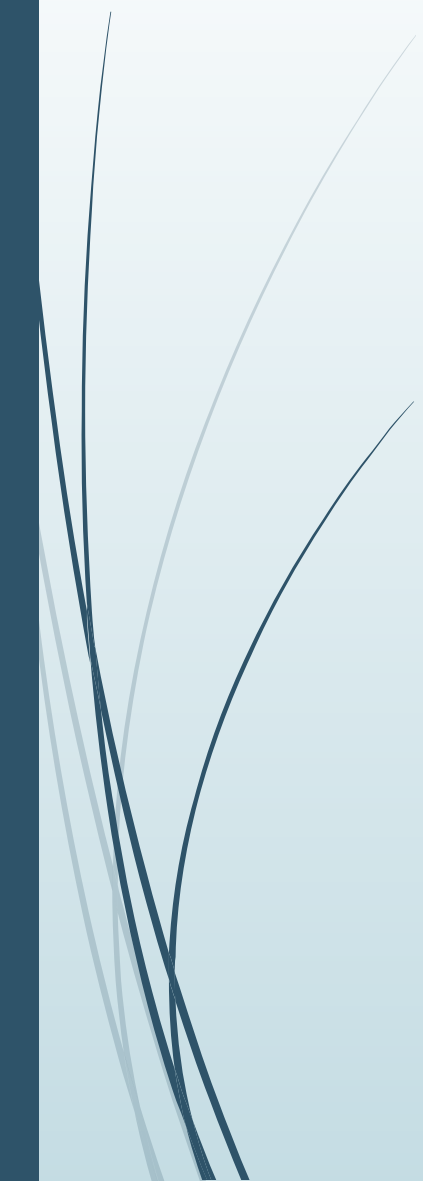
4. **Performance Measure** — How Well the Agent Performs

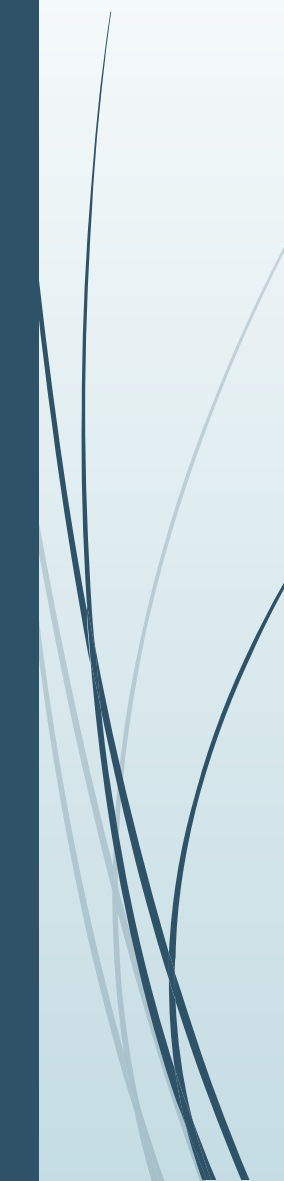
This tells us **how efficiently** the agent is achieving its goal.

We evaluate it based on the amount of dirt cleaned, the time taken, and how effectively both rooms are cleaned.



Problem Solving Agents

- Input-----output
 - Percept-----action
 - Problem-----solution
- 



AI Agent	Percepts (Inputs Sensed)	Actions (Outputs Executed)	Target Goal (Objective)	Performance Measures (Success Metrics)
Google Maps	• GPS coordinate streams • Live speed/congestion data • User hazard/delay reports • Historic travel pattern records	• Render updated map visualization • Play audio turn-by-turn prompts • Recalculate dynamic ETA metrics	Achieve optimal path routing to destination.	• Absolute travel time minimized • Distance traveled minimized • Toll/traffic exposure minimized



PEAS

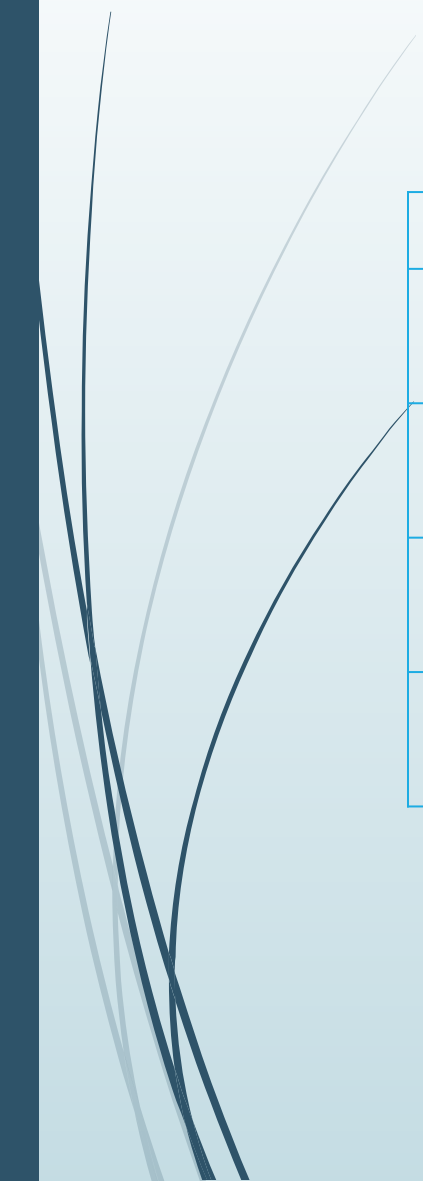
- Specifying the task environment is always the first step in designing agent
- PEAS:
 - Performance
 - Environment
 - Actuators
 - Sensors

Taxi Driver Example

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard



E.g. Automated taxi driver:



PEAS	Specification
Performance	Safe, fast, legal, comfortable trip; maximize profits
Environment	Roads, other traffic, pedestrians, customers
Actuators	Steering wheel, accelerator, brake, signal, horn
Sensors	Cameras, sonar, speedometer, GPS, odometer, speech input



Medical diagnosis system:

PEAS	Specification
Performance	Healthy patient, minimize cost/liability
Environment	Patient, hospital, staff
Actuators	Screen display, tests, treatments
Sensors	Keyboard (symptom entry), test results

Examples of agent types

Agent Type	P	E	A	S
Medical Diagnosis	Healthy patient, minimize costs, lawsuits	Patient, hospital, staff	Display questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, test results, patient's answers
Satellite image system	Correct image categorization	Downlink from satellite	Display categorization of scene	Color pixel array
Interactive English tutor	Maximize student's score on test	Set of students, testing agency	Display exercises, suggestions, corrections	Keyboard entry

94 Summary

- An **agent is something that perceives and acts in an environment. The agent function**
- for an agent specifies the action taken by the agent in response to any percept sequence.
- • The **performance measure evaluates the behavior of the agent in an environment. A rational agent acts so as to maximize the expected value of the performance measure**, given the percept sequence it has seen so far.
- • A **task environment specification includes the performance measure, the external environment**, the actuators, and the sensors. In designing an agent, the first step must always be to specify the task environment as fully as possible.
- • Task environments vary along several significant dimensions. They can be fully or partially observable, single-agent or multiagent, deterministic or stochastic, episodic or sequential, static or dynamic, discrete or continuous, and known or unknown.

Case Study: Working of an Intelligent Agent

□ The Vacuum Cleaner Agent.

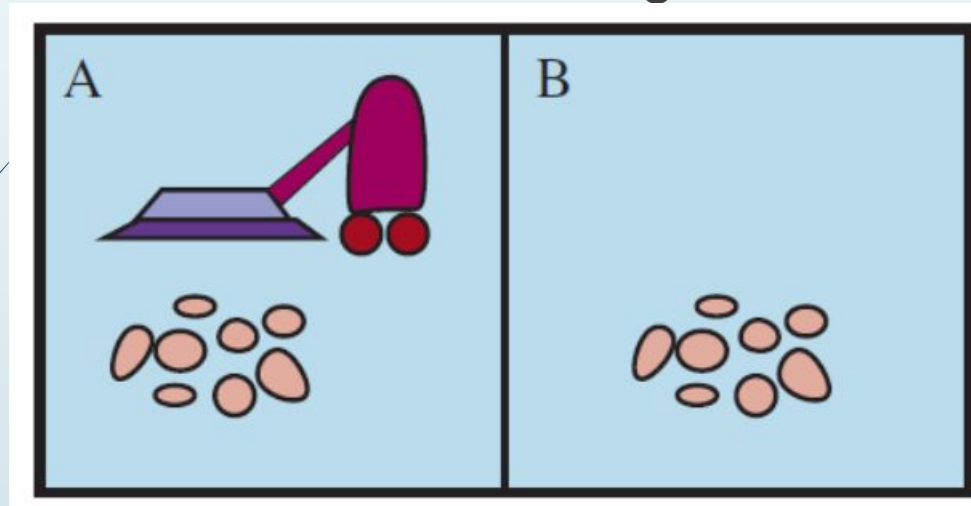


Figure 2 : Vacuum Cleaner Agent in a Two-Room Environment

The Environment

Imagine two rooms — **Room A** and **Room B**. Each room can be either:

 **Clean** or  **Dirty**

The agent (our vacuum cleaner robot) can:
Move between the two rooms
Clean when needed

The Agent System

The Vacuum Cleaner Agent consists of four main components:

1. **Environment**: Two locations — A and B.
2. **Sensors**: Detect if the current room is **clean** or **dirty**.
3. **Agent (Decision Maker)**: Decides what to do next.
4. **Effectors (Actions)**: Perform tasks like clean or **Move** (Left/Right).
5. **How It Works**

Situation	Action
In A and Dirty	Clean
In A and Clean	Move Right
In B and Dirty	Clean
In B and Clean	Move Left



Agent Function:

Percept Sequence	Action
[A, Clean]	Move Right
[A, Dirty]	CLEAN
[B, Clean]	Move Left
[B, Dirty]	CLEAN
[A, Clean], [A, Clean]	Move Right
[A, Clean], [A, Dirty]	CLEAN

*Note:
infinite*



Agent Program: actual code or algorithm

(Python logic example)

```
if status == "Dirty":  
    action = "CLEAN"  
elif location == "A":  
    action = "Right"  
elif location == "B":  
    action = "Left"
```



Summary table of Agent Function vs. Agent Program

Concept	Meaning	Example
Agent Function	The mathematical idea — it tells what the agent should do for every possible percept sequence.	“If dirty $\rightarrow$ clean; if clean $\rightarrow$ move”
Agent Program	The actual implementation (code or algorithm) that runs inside a machine.	The software inside the robot that follows those rules

Example 1 vacuum Cleaner

- **States:** The agent is in one of two locations, each of which might or might not contain dirt. Thus there are $2 \times 2^2 = 8$ possible world states.
- **Initial state:** Any state can be designated as the initial state.
- **Successor function:** This generates the legal states that result from trying the three actions (Left, Right, and Suck).
- **Goal test:** This checks whether all the squares are clean.
- **Path cost:** Each step costs 1, so the path cost is the number of steps in the path.

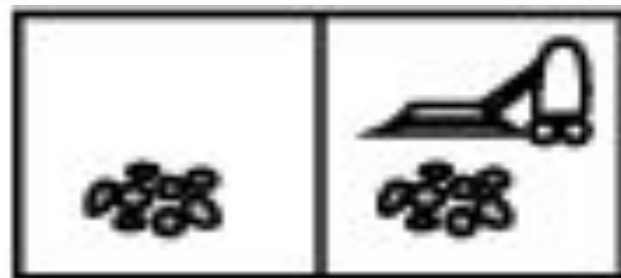
Vacuum Cleaner World

- State space(Successor Function) = 8
 - Action = 3 i. e left right and suck
 - Goal = to clean all dirt(states 7 and 8)
 - If the environment is observable, deterministic,
and completely known, then the problem is trivially solvable by any of the algorithms.
 - For example, if the initial state is 5, then the action sequence [Right , Suck] will reach a goal state, 8.
- >>>> continue diagram

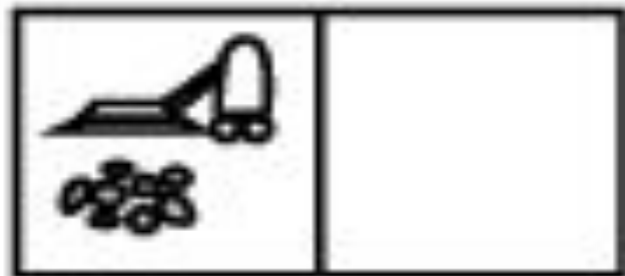
1



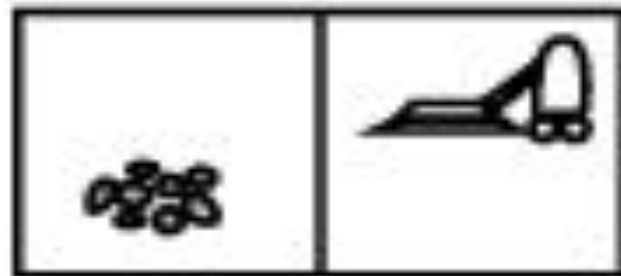
2



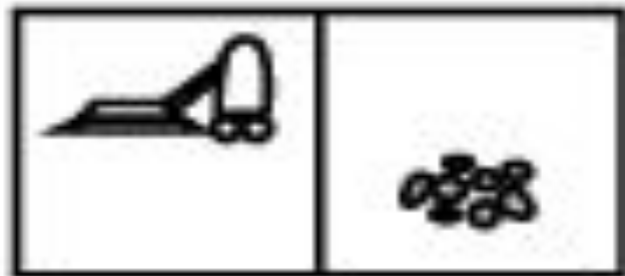
3



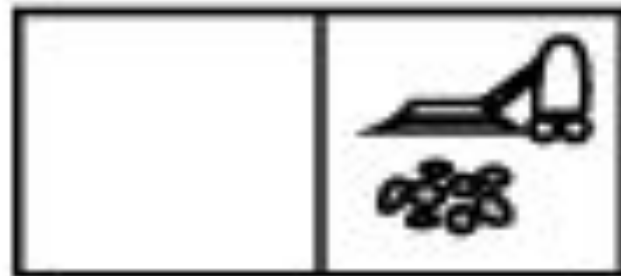
4



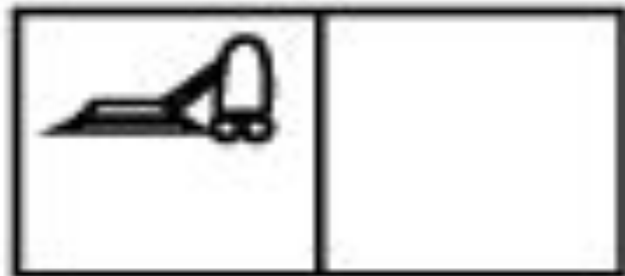
5



6



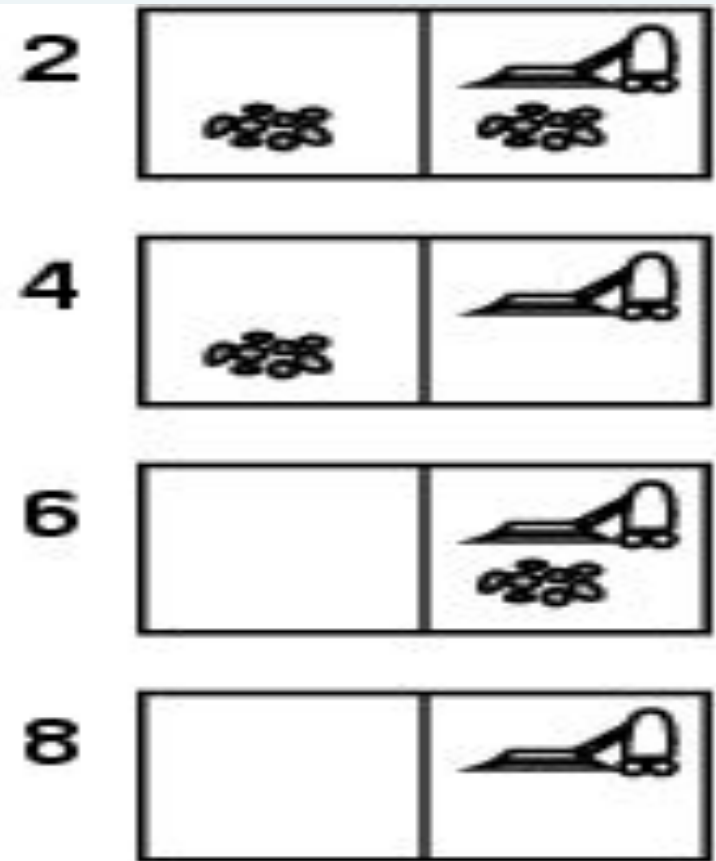
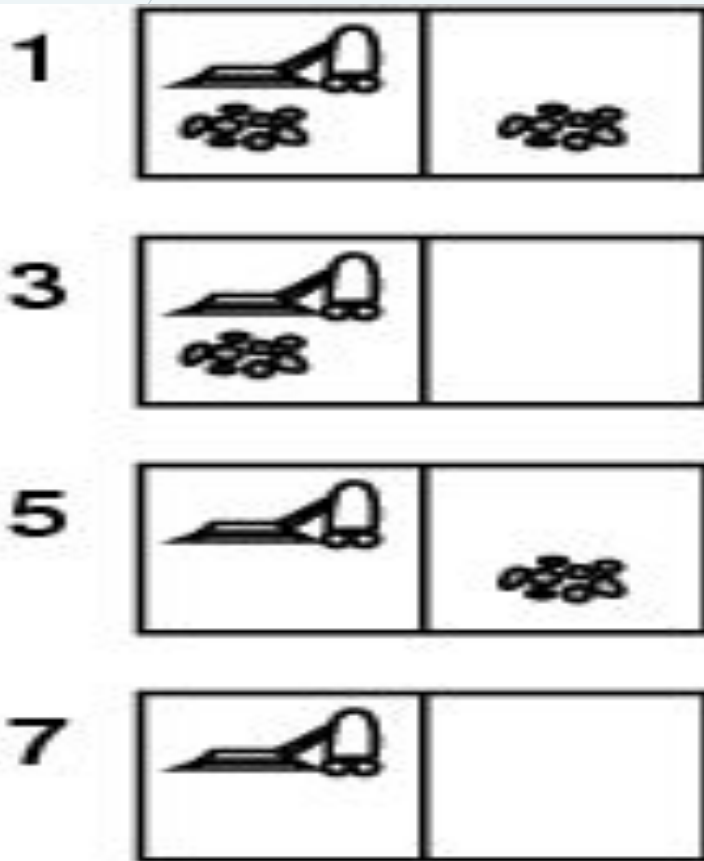
7



8

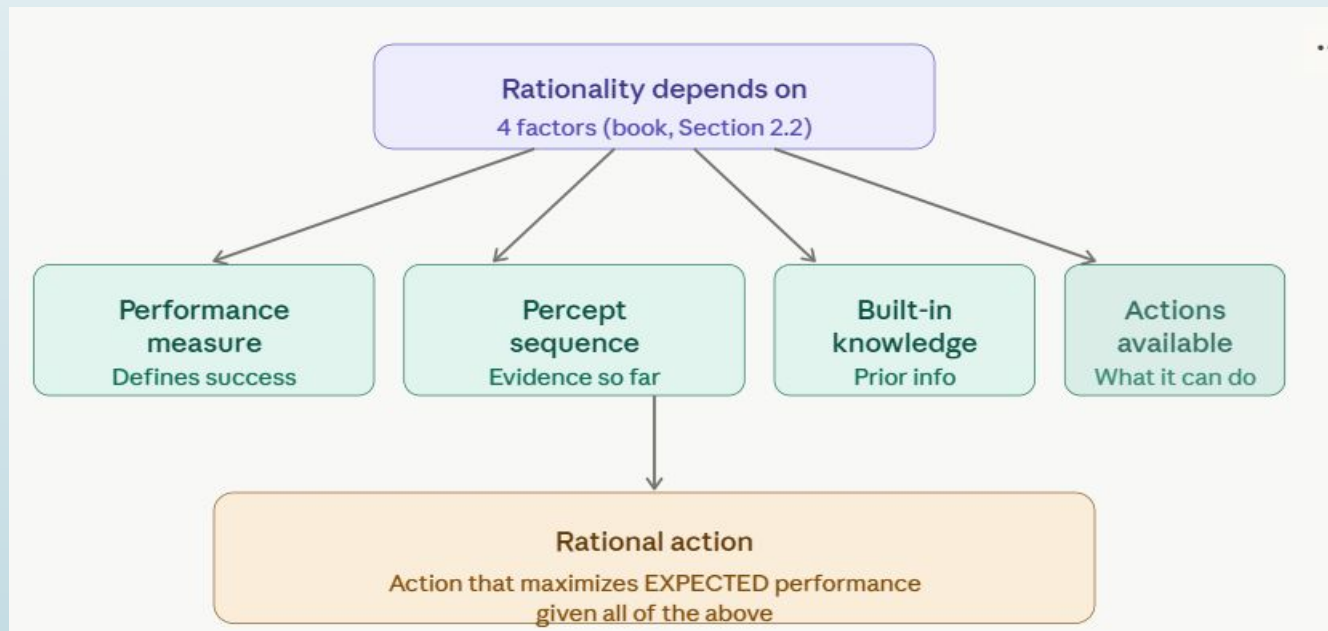


Single-state, start in #5. suppose
Solution? Sequence of action=[Right,
CLEAN] Goal state = 8



Rationality

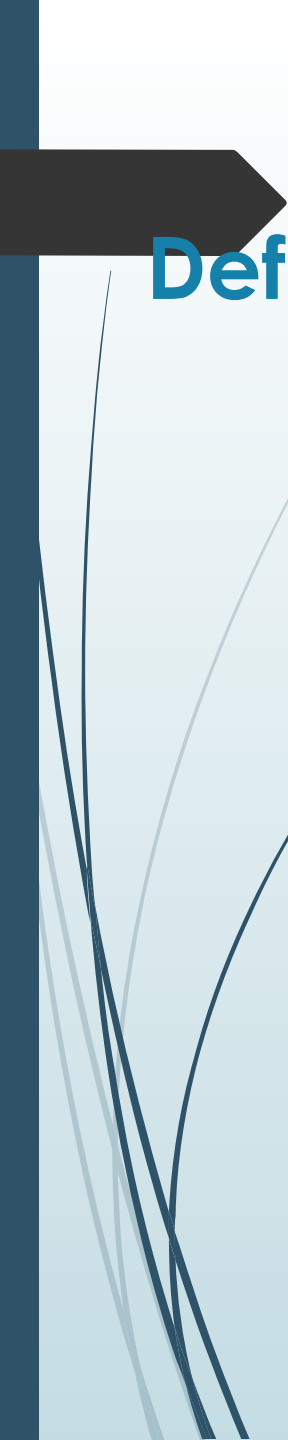
- Rational Agent-Does Write Things
- "As a general rule, it is better to design performance measures according to what one actually wants in the environment, rather than according to how one thinks the agent should behave





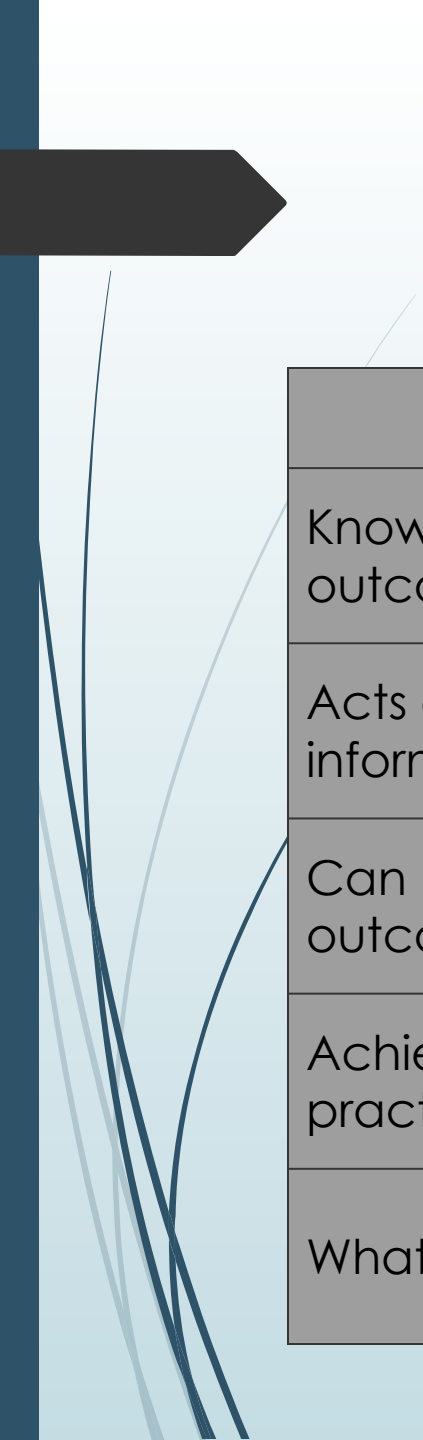
Good Behavior: Concept of Rationality

- A performance measure embodies the criterion for success of an agent's behavior
- E.g. "amount of dirt cleaned up in an 8-hour shift"
- For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.
- **Four factors that determine rationality :**
 - The performance measure that defines the criterion of success
 - The agent's prior knowledge of the environment
 - The actions that the agent can perform
 - The agent's percept sequence to date
- **Rationality is NOT perfection.** Rationality maximizes *expected* performance. Perfection maximizes *actual* performance
- Rationality... requires actions that gather useful information... an agent that fails to gather information when it could is not rational.



Definition of a rational agent:

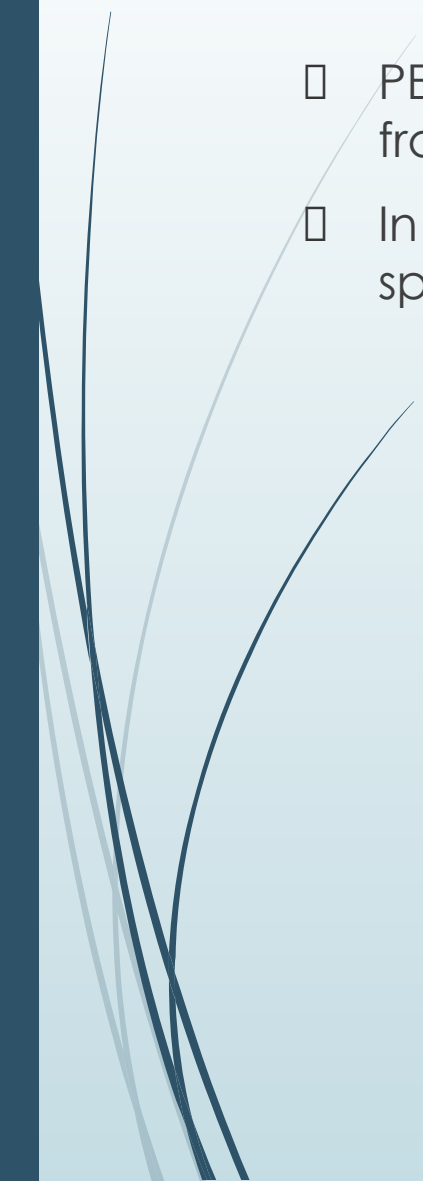
- *For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.*



	Omniscient Agent	Rational Agent
Knows actual outcomes	Yes	No
Acts on available information	Yes	Yes
Can produce bad outcomes	No	Yes (if unlucky)
Achievable in practice	No	Yes
What it maximizes	Actual performance	Expected performance



The Nature of Environments

- PEAS (Performance, Environment, Actuators, Sensors): framework for specifying the **task environment**
 - In designing an agent, the first step must always be to specify the task environment as fully as possible
- 

Environment

- **the environment is everything outside the agent that the agent interacts with.**
- The agent cannot exist without an environment. Every agent is always situated *inside* some environment, perceiving it and acting on it.
- An agent perceives its **environment** through sensors and acts upon that environment through actuators.
- E.g.

Agent	Its Environment
Vacuum cleaner robot	Rooms, floors, dirt, furniture, walls
Chess-playing program	The chessboard, opponent's moves, clock

Properties of task Environments

□ Fully-observable vs. Partially-observable

- If an agent's sensors give it access to the complete state of the environment at each point in time, then the task is fully-observable. Example: Automated Taxi can not see what other drivers are thinking □ Partially observable
- E.g Sudoku –everything is accessible no hidden rounds
- Online exam system- tells students only questions and time

□ Deterministic vs. Stochastic

- If the next state of the environment is completely determined by the current state and the action executed by the agent, then the environment is deterministic. Example: Taxi driving is stochastic.
- **Strategic**: deterministic except for the actions of other agents
- E.g. Calculator- for any no. of times gives certain answers
- Travel time, rain-uncertain

□ Episodic vs. Sequential

- The agent's experience is divided into atomic episodes. Each episode consists of the agent perceiving and then performing a single action.
 - E.g MCQ exam- current decision is not future decision dependent
 - Exam preparation

Properties of task Environments ..cont.

□ **Static vs. Dynamic**

- Environment remains unchanged Vs changed
- If the environment can change while an agent is deliberating, then the environment is dynamic.
- **Semidynamic**: the agent's performance score changes only.
- E.g. Crossword
- Taxi driver system

□ **Discrete vs. Continuous**

- Discrete- Finite no. of states
- E.g Student attendance
- Continuous-infinite no. of possibilities
- E.g. Writing speed of a student

□ **Single agent vs. Multiagent**

- E.g. Practice Coding alone
- Miniproject, Group Project

Properties of Task Environments:

1. Fully observable vs. Partially observable:

If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable."

A fully observable environment is convenient — the agent needs no internal state to keep track of the world. Partial observability occurs because of noisy or inaccurate sensors, or because parts of the state are simply absent from the sensor data.

2. Single agent vs. Multiagent:

-*competitive* multiagent environments (chess: maximizing your performance minimizes the opponent's)

-*cooperative* ones (multiple robots in a warehouse).

"The agent-design problems in multiagent environments are often quite different from those in single-agent environments."

3. Deterministic vs. Stochastic:

"If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic."
if an environment is deterministic except for the actions of other agents, it is called **strategic**.

4. Episodic vs. Sequential:

"In an episodic task environment, the agent's experience is divided into atomic episodes... the choice of action in each episode depends only on the episode itself."

"In sequential environments, the current decision could affect all future decisions."
e.g Chess and taxi driving are sequential. Image classification is episodic.



5. Static vs. Dynamic:

"If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent; otherwise it is static."

A **semidynamic** environment—the environment itself doesn't change but the agent's performance score does — time-limited chess is an example.

6. Discrete vs. Continuous:

"The discrete/continuous distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agent."

Chess has a discrete, finite set of states. Taxi driving has continuous time, speed, and position.

7. Known vs. Unknown:

"In a known environment, the outcomes... for all actions are given. If the environment is unknown, the agent will have to learn how it works in order to make good decisions."

Consider the automated taxi

- the visual input from a single camera comes in at the rate of roughly 27 megabytes per second (30 frames per second, 640×480 pixels with 24 bits of color information).
- This gives a lookup table with over $10^{250,000,000,000}$ entries for an hour's driving.

Characterizing a Task Environment

- Must first specify the setting for intelligent agent design.
- **PEAS: Performance measure, Environment, Actuators, Sensors**
- **Example:** the task of designing a self-driving car
 - **Performance measure** Safe, fast, legal, comfortable trip
 - **Environment** Roads, other traffic, pedestrians
 - **Actuators** Steering wheel, accelerator, brake, signal, horn
 - **Sensors** Cameras, LIDAR (light/radar), speedometer, GPS, odometer engine sensors, keyboard



- **1) Fully observable / Partially observable**
 - If an agent's sensors give it access to the **complete state of the environment** needed to choose an action, the environment is **fully observable**.
(e.g. chess)



2) Deterministic / Stochastic

- An environment is **deterministic** if the next state of the environment is completely determined by the current state of the environment and the action of the agent;
- In a **stochastic** environment, there are multiple, unpredictable outcomes. (If the environment is deterministic except for the actions of other agents, then the environment is **strategic**).

In a fully observable, deterministic environment, the agent need not deal with uncertainty.

Note: Uncertainty can also arise because of computational limitations. E.g., we may be playing an **omniscient** (“all knowing”) opponent but we may not be able to compute his/her moves.

□ 3) Episodic / Sequential

In an **episodic** environment, the agent's experience is divided into atomic episodes. Each **episode** consists of the agent perceiving and then performing a single action.

- Subsequent episodes do not depend on what actions occurred in previous episodes. Choice of action in each episode depends only on the episode itself. (E.g., classifying images.)
- In a **sequential** environment, the agent engages in a series of connected episodes. Current decision can affect future decisions. (E.g., chess and driving)

□ 4) Static / Dynamic

- A **static** environment does not change while the agent is thinking.
- The passage of time as an agent deliberates is irrelevant.
- The environment is **semidynamic** if the environment itself does not change with the passage of time but the agent's performance score does.

5) Discrete / Continuous

- If the number of distinct percepts and actions is limited, the environment is **discrete**, otherwise it is **continuous**.

6) Single agent / Multi-agent

- If the **environment contains other intelligent agents**, the agent needs to be concerned about strategic, game-theoretic aspects of the environment (for either cooperative or competitive agents).
- Most **engineering environments** don't have multi-agent properties, whereas most **social and economic systems** get their complexity from the interactions of (more or less) rational agents.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle Chess with a clock						
Poker Backgammon						
Taxi driving Medical diagnosis						
Image analysis Part-picking robot						
Refinery controller Interactive English tutor						

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

Examples of Task Environments and Characteristics

Task environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
"The hardest case"	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete
English dynamic controller	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

STRUCTURE OF AGENTS

- *behavior—the action that is performed after any given sequence of percepts. Now we must bite the bullet and talk about how the insides work.*
- The job of AI is to design an **agent program that implements the agent function**—the mapping from percepts to actions.
 - *agent = architecture + program .*

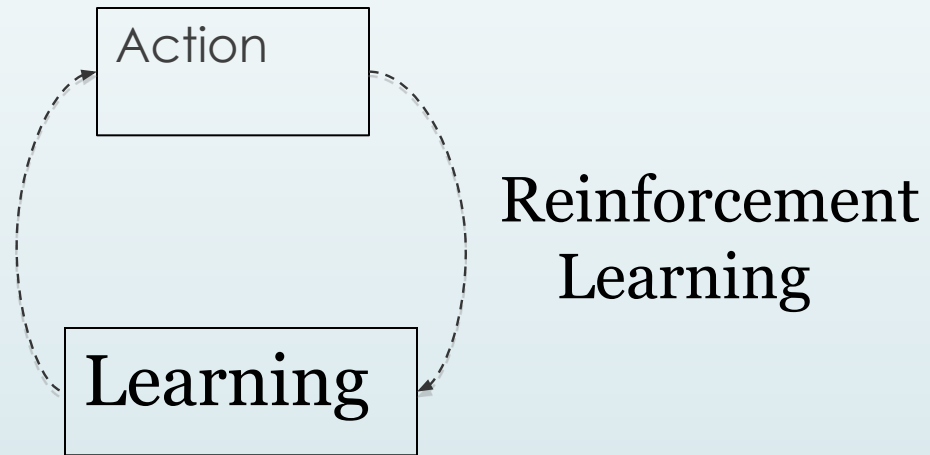


four basic kinds of agent programs

□ embody the principles underlying almost all intelligent systems:

- A. • Simple reflex agents;
- B. • Model-based reflex agents;
- C. • Goal-based agents;
- D. • Utility-based agents.

Reflex-Based Agents



- Studied in AI, Cybernetics, Control Theory, Biology, Psychology.

A Reflex-Based Agent chooses its action based only on the **current percept (current situation)**.

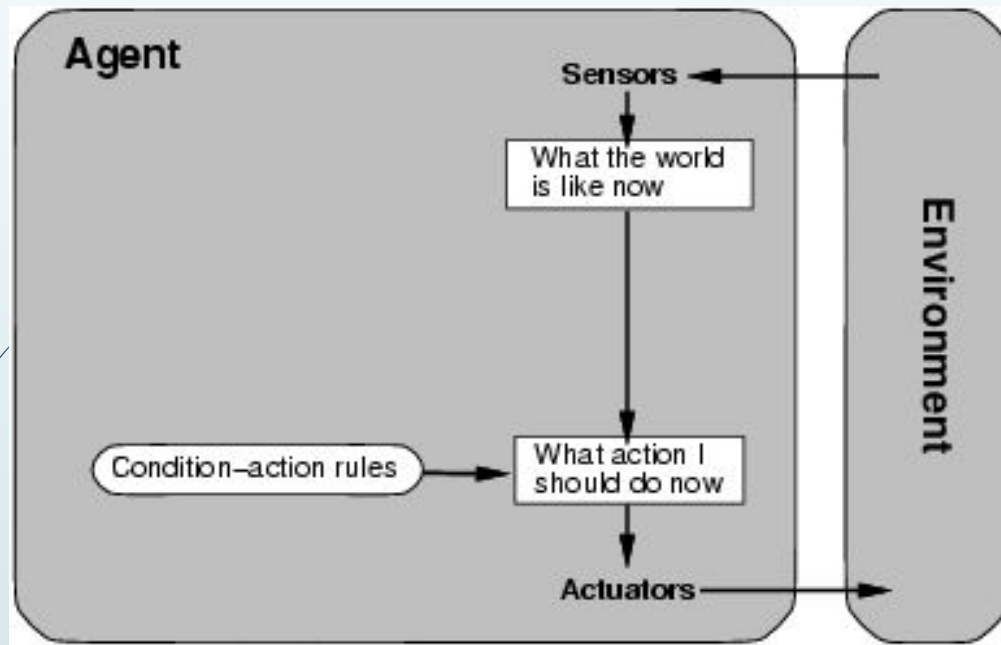
It does not remember the past and does not predict the future.



Simple Reflex Agents

- Agents choose actions only based on the **current percept**.
- They are rational only if a correct decision is made only on the basis of current precept.
- Their environment is completely observable.
- Fully Observable environment (properly working on available conditions)
- E.g. Temperature sensing systems (e.g. AC-switch on/off)

Simple reflex agents



Simple reflex agents

- Reflex-----quick
- Fully observable
- Depends on current situation given by sensors, then actuators will perform action on environment.

- Limitations of Simple reflex agents

Simple but very limited intelligence.

Action does not depend on percept history, only on current percept. Therefore no memory requirements.

- Infinite loops

e.g.

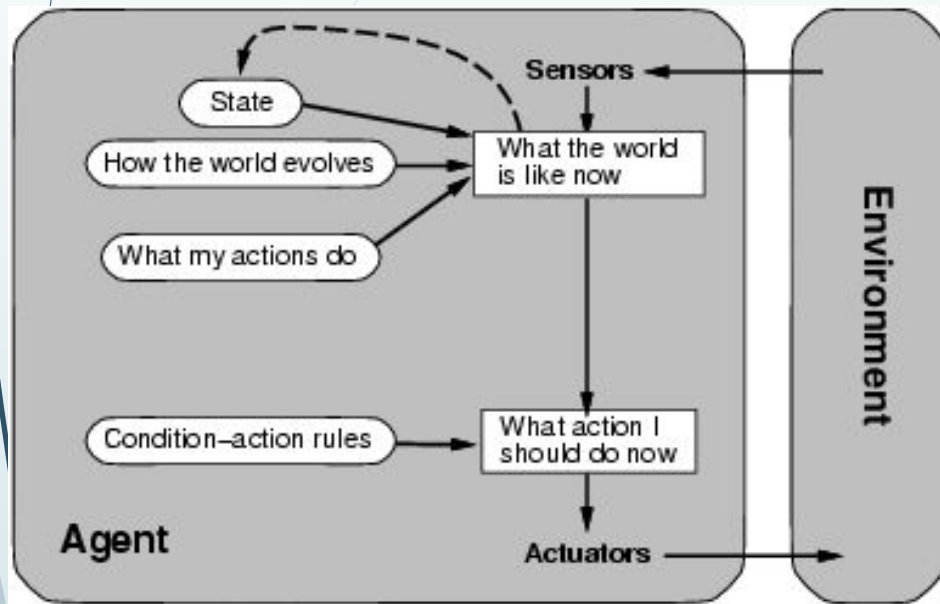
Suppose vacuum cleaner does not observe location. What do you do given location = clean?
Left of A or right on B -> infinite loop.

Fly buzzing around window or light.

Possible Solution: Randomize action.

Chess – openings, endings (Lookup table)

Model-based reflex agents



```
function REFLEX-AGENT-WITH-STATE(percept) returns action
  static: state, a description of the current world state
         rules, a set of condition-action rules

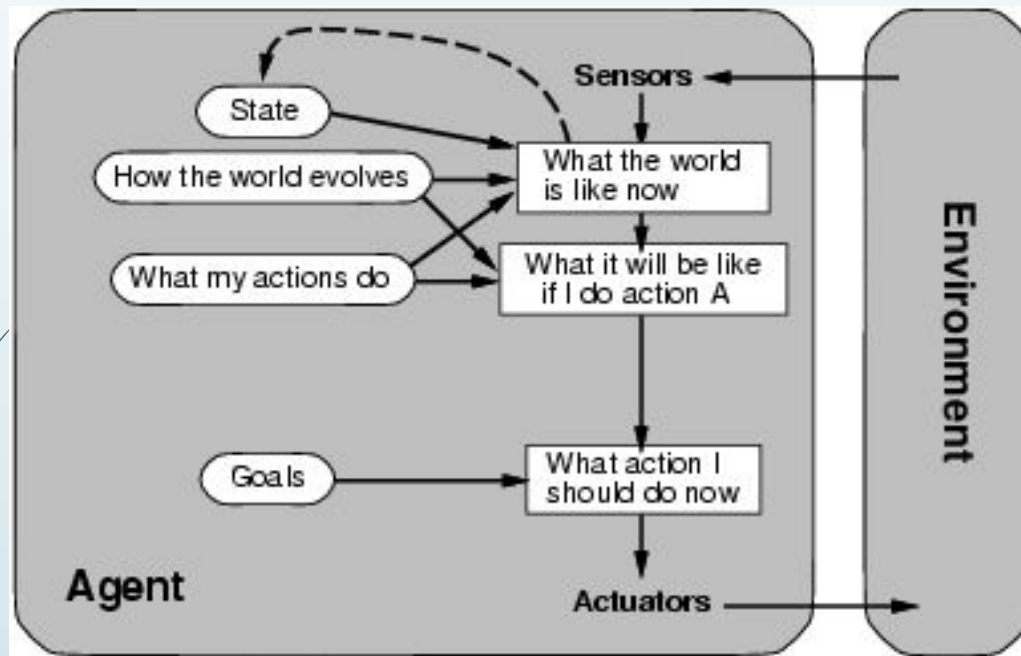
  state ← UPDATE-STATE(state, percept)
  rule ← RULE-MATCH(state, rules)
  action ← RULE-ACTION[rule]
  state ← UPDATE-STATE(state, action)
  return action
```

- Works by finding the rule whose condition matches **current situation**. (e.g. classroom envir.)
- Can work in **partially observable** env. And track situation.
- **Stores percept history**
- Agent keeps track of internal state which is adjusted by each percept and that depends on percept history.
- **Knowledge Based**
e.g Overtaking car gets closer from behind based on previous knowledge
- **How agents actions affect the world**
 - Wheel turned clockwise takes you right
- Model base agents update their state

Goal-based agents

- Expansion of Model Based Agent
- Reflex Agent know desirable situation i.e. goal
- Focuses only on reaching goal set.
- Agent takes decision based on how far it is currently from the goal state.
- Every action is taken to minimize distance to goal state.
- More flexible
- knowing state and environment? Enough?
- Have a goal A destination to get to
- Uses knowledge about a goal to guide its actions
 - E.g., Search, planning

Goal-based agents

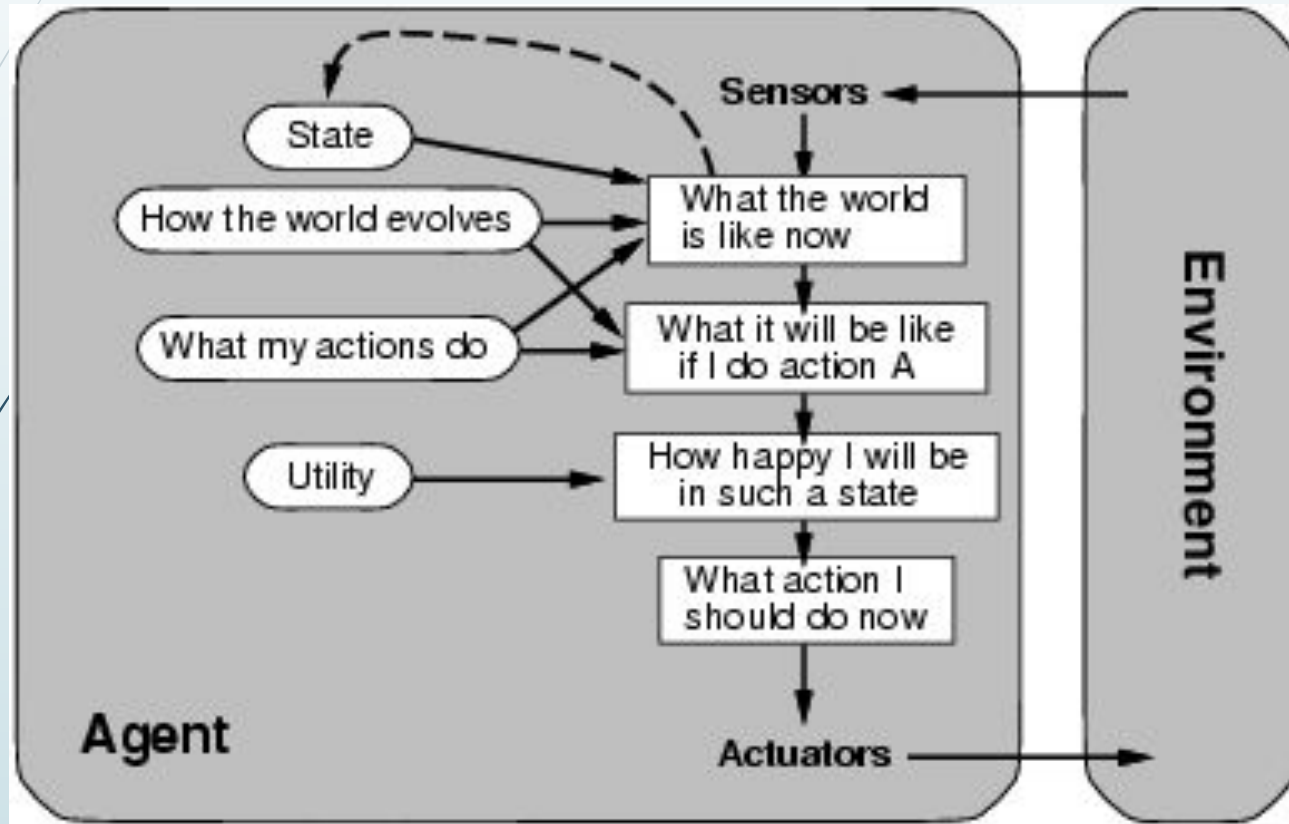


- Every time agent checks are you going closer to your goal or not for every action.
- E.g. robots designed to deliver packets for e commerce web site

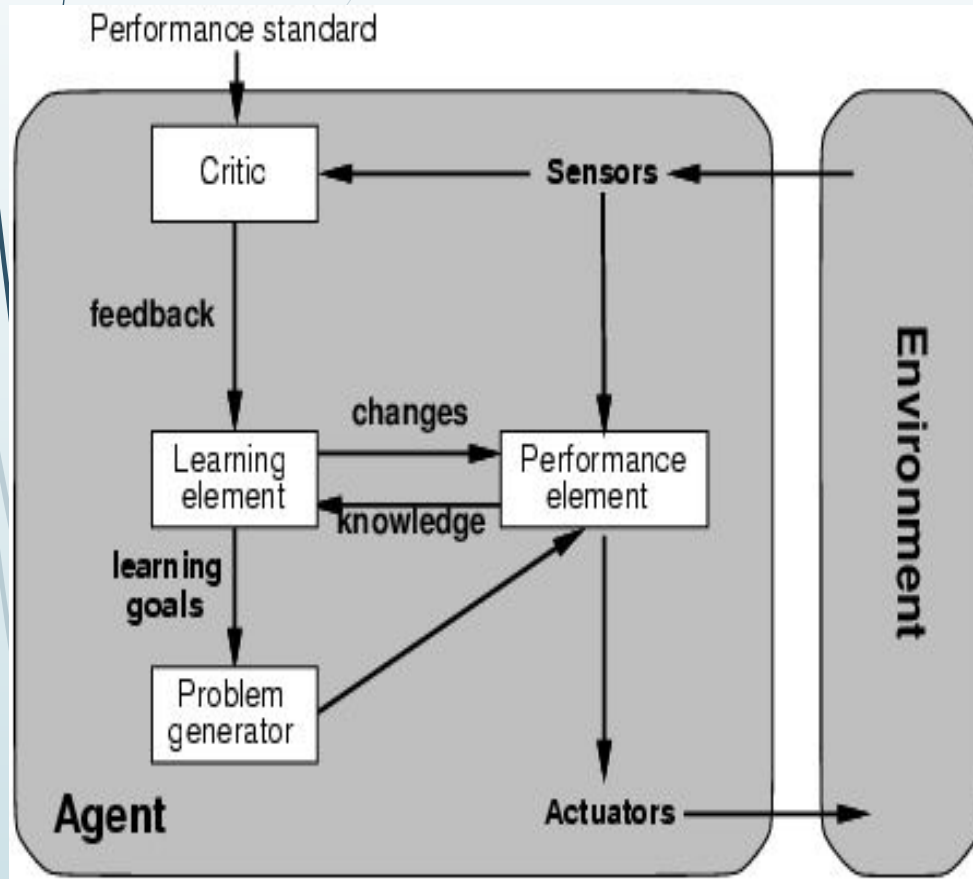
Utility-based agents

- Goals are not always enough
 - Many action sequences get taxi to destination
 - Consider other things. How fast, how safe.....
- Focus is not goal (destination) but on utility function (add a feature of state of life)
- Partially observable environment
- Uses advancement in goal function with utility
- A utility function maps a state onto a real number which describes the associated degree of **“happiness”, “goodness”, “success”**.
- Where does the utility measure come from
 - Economics: money.
 - Your life state-satisfaction,happiness
- E.g GPS shows short term path, but selected path is blocked. It is unhappy situation, then GPS shows another route with less traffic for reaching destination, which gives happy state goal.

Utility-based agents

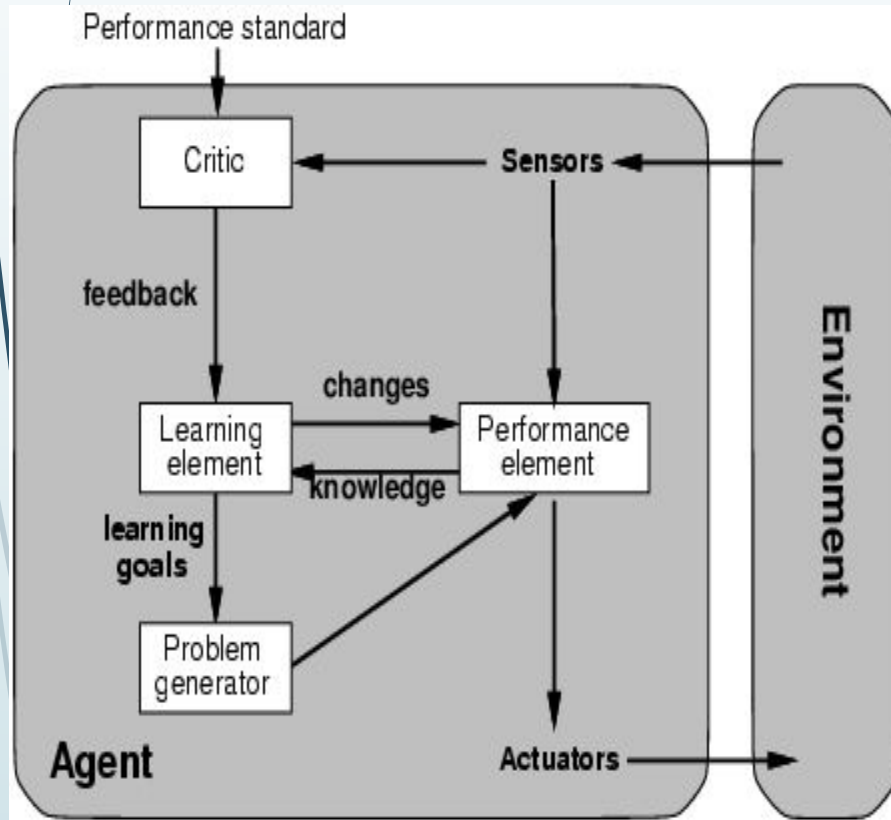


Learning agents



- Performance element- -
 - decides how to do?
 - Based on input sensor decides output action
- Learning element-
 - Enables the agent to learn things and Modifies performance element.
 - When and what to do?
- Problem generator
 - Tries to solve the problem differently instead of optimizing.
 - Suggests **exploring** new actions -> new problems.

Learning agents



● Critic-

- Tries to reduce errors
- Take input through environment directly
- how the agent is doing
 - Input
 - Fixed

Learning agents(Taxi driver)

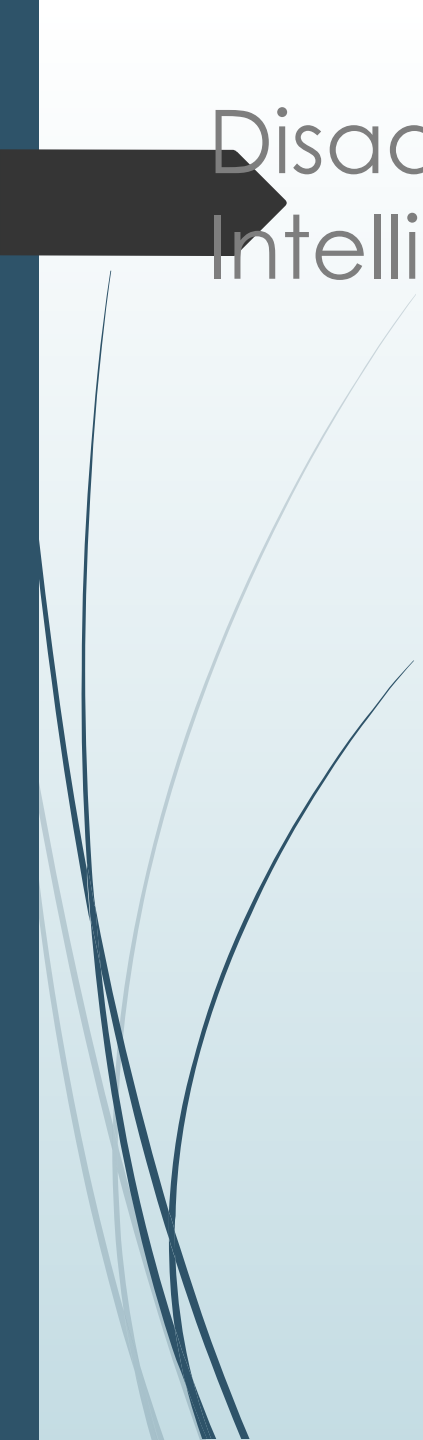
- Performance element
 - How it currently drives
- Taxi driver Makes quick left turn across 3 lanes
 - Critics observe shocking language by passenger and other drivers and informs bad action
 - Learning element tries to modify performance elements for future
 - Problem generator suggests experiment out something called Brakes on different Road conditions
- Exploration vs. Exploitation
 - Learning experience can be costly in the short run
 - shocking language from other drivers
 - Less tip
 - Fewer passengers

13 8 Summary

- • The **agent program implements the agent function. There exists a variety of basic** agent-program designs reflecting the kind of information made explicit and used in the decision process. The designs vary in efficiency, compactness, and flexibility. The appropriate design of the agent program depends on the nature of the environment.
- • **Simple reflex agents respond directly to percepts, whereas model-based reflex agents**
- maintain internal state to track aspects of the world that are not evident in the current percept. **Goal-based agents act to achieve their goals, and utility-based agents try to** maximize their own expected “happiness.”
- • *All agents can improve their performance through **learning**.*

Advantages of Artificial Intelligence

- More powerful and more useful computers
- Conversion of information into knowledge
- New and improved interfaces
- Solving new problems
- Better handling of information
- Relieves information overload



Disadvantages of Artificial Intelligence

- ▣ Increased costs
- ▣ Difficulty with software development - slow and expensive
- ▣ Few experienced programmers
- ▣ Few practical products have reached the market as yet.

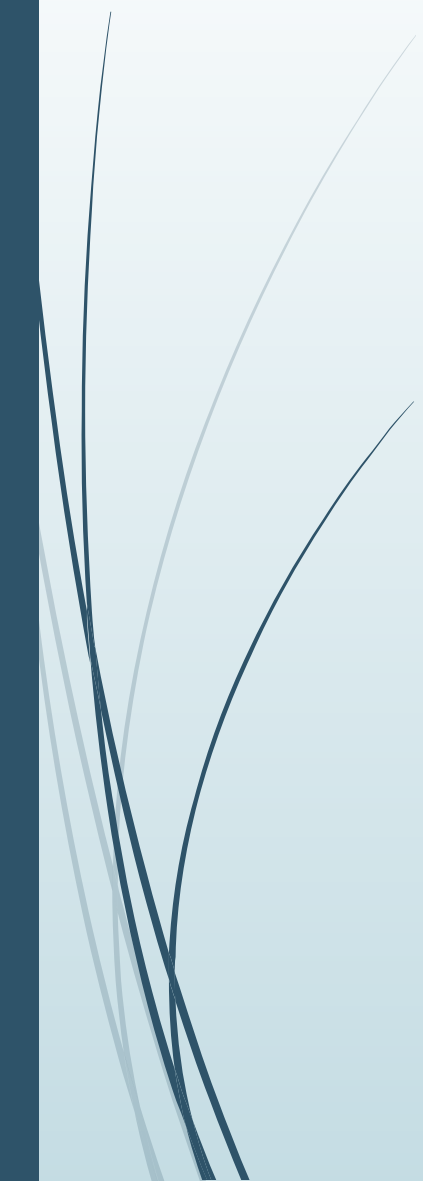


Problem-Solving

- ▣ **Problem characteristics**
- ▣ **Production systems**
- ▣ **control strategies**



Problem-Solving



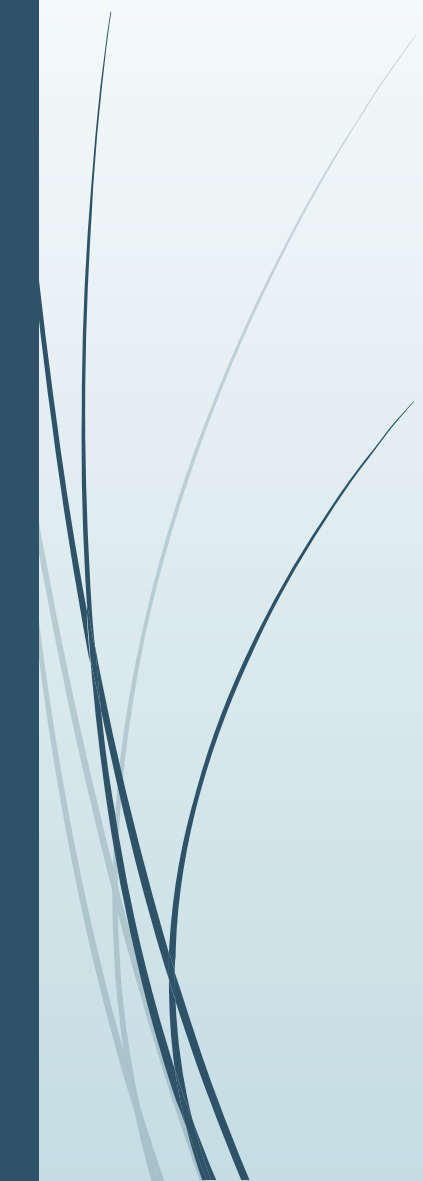


Search Techniques

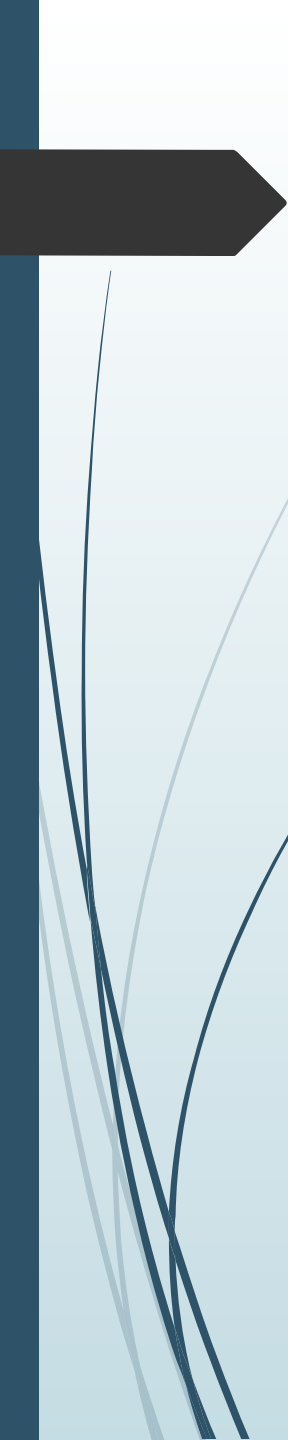
- **Uninformed search (breadth-first, depth-first)**
- **informed/heuristic search (hill climbing, best first)**
- **heuristic functions.**



References



1. Book
2. <https://aiknowledgehub.in/fundamentals-of-ai>
3. https://study.iitm.ac.in/ds/course_pa...
4. course on "AI: Search Methods for Problem Solving" by Prof. Deepak Khemani
5. *Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd Edition, Prentice Hall, 2009.*
6. *Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India), 2013. (Chapters 1 – 8, some parts from Chapters 9 and 10))*



Thank You

AI Agent	Percepts (Inputs Sensed)	Actions (Outputs Executed)	Target Goal (Objective)	Performance Measures (Success Metrics)
Google Maps	• GPS coordinate streams • Live speed/congestion data • User hazard/delay reports • Historic travel pattern records	• Render updated map visualization • Play audio turn-by-turn prompts • Recalculate dynamic ETA metrics	Achieve optimal path routing to destination.	• Absolute travel time minimized • Distance traveled minimized • Toll/traffic exposure minimized
Self-Driving Car	• Camera pixel matrix arrays • LiDAR spatial point clouds • Echo radar frequency waveforms • Speedometer & inertial readings	• Rotate steering wheel actuator • Engage brake pad hydraulics • Adjust electronic throttle valve • Toggle lane signal illumination	Transport passengers safely to destination coordinates.	• Zero safety collisions/infractions • Adherence to local traffic rules • Passenger physical comfort index • Speed optimization
ChatGPT	• Input text token sequences • Voice microphone audio waveforms • Uploaded multi-media file data • Active session context history	• Stream output text response tokens • Play synthesized voice stream • Invoke background application APIs • Generate structured data blocks	Assist human requests and streamline communication.	• Contextual prompt relevance • Textual/factual accuracy score • Response latency (speed) • End-user satisfaction feedback

□ 1.2.2 Mathematics

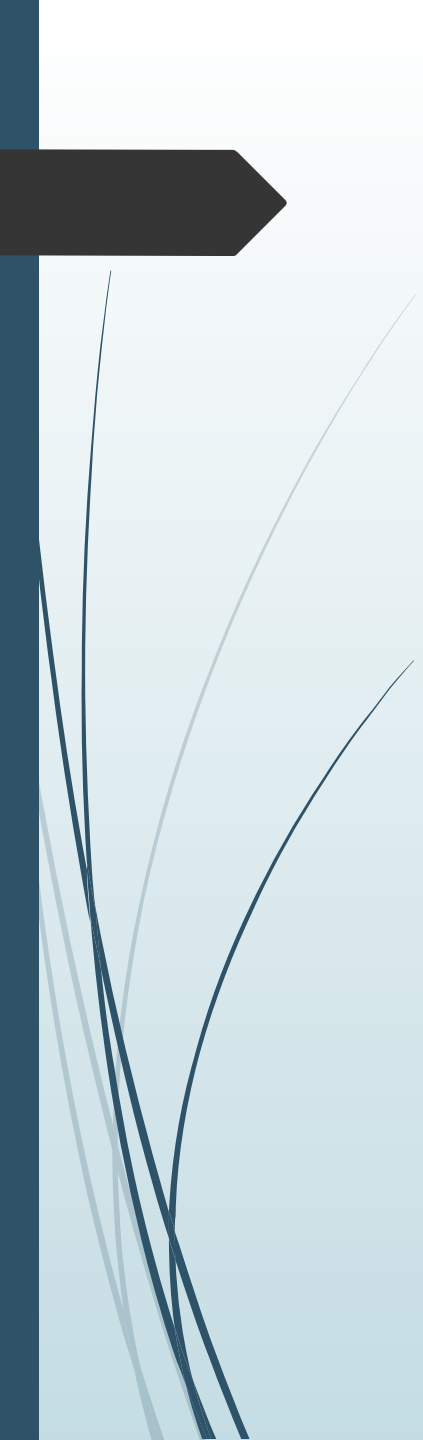
1. What are the formal rules to draw valid conclusions?
2. What can be computed?
3. How do we reason with uncertain information?

□ Three fundamental areas: logic, computation, and probability.

Mathematics

□ Logic

- George Boole (1815-1864): formal language for making logical inference
- Gottlob Frege (1848-1925): First-order logic (FOL)
- Computability
 - David Hilbert (1862-1943): Problem #23: is there an algorithm for deciding the truth of any logical proposition involving the natural numbers?
 - Kurt Godel (1906-1978): No: undecidability (yes for FOL)
 - Alan Turing (1912-1954): which functions are computable?
 - Church-Turing thesis: any computable function is computable via a Turing machine
 - No machine can tell in general whether a given program will return an answer on a given input, or run forever



Mathematics...

- Intractability
 - Polynomial vs. exponential (Cobham 1964; Edmonds 1965)
 - Reduction (Dantzig 1960, Edmonds 1962)
 - NP-completeness (Steven Cook 1971, Richard Karp 1972)
 - Contrasts “Electronic Super-Brain”

Mathematics...

- Probability
 - Gerolamo Cardano (1501-1576): probability in gambling
 - Pierre Fermat (1601-1665), Blaise Pascal (1623-1662), James Bernoulli (1654-1705), Pierre Laplace (1749-1827): new methods
 - Bernoulli: subjective beliefs->updating
 - Thomas Bayes (1702-1761): updating rule
- Decision theory = probability theory + utility theory
 - John Von Neumann & Oskar Morgenstern 1944
- Game theory

(Boole, 1847).

worked out the details of propositional, or Boolean, logic

Gottlob
Frege
(1848–1925)

extended Boole's logic to include objects and relations, creating the first order logic that is used today.

The next step was to determine the limits of what could be done with logic and computation. The first ALGORITHM nontrivial algorithm

Alan Turing
(1912–1954)

to characterize exactly which functions are **computable—capable of being computed.**

The Italian
PROBABILITY
Gerolamo
Cardano
(1501–1576)

Besides logic and computation, the third great contribution of mathematics to AI is the theory of **probability.**

□ 1.2.3 Economics

How should we make decisions so as to maximize payoff?

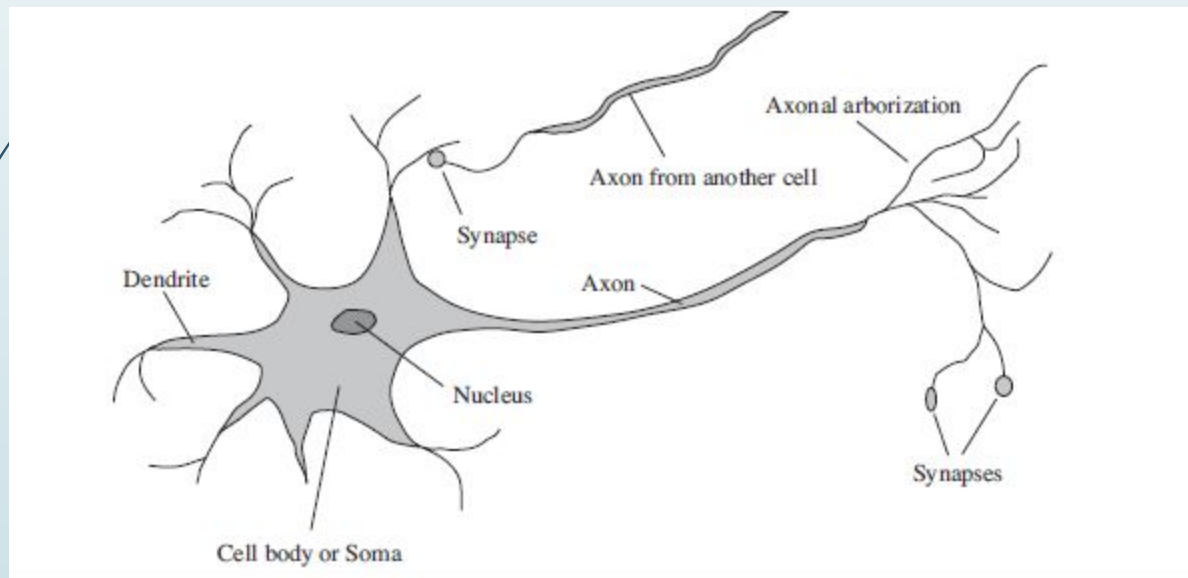
2. How should we do this when others may not go along?
3. How should we do this when the payoff may be far in the future?

□ *Most people think of economics as being about money, but economists will say that they are really studying how people make choices that lead to preferred outcomes.*

□ 1.2.4 Neuroscience

How do brains process information?

- **Neuroscience is the study of the nervous system, particularly the brain.**



□ 1.2.5 Psychology

How do humans and animals think and act?

- The origins of scientific psychology are usually traced.
- **Cognitive psychology, which views COGNITIVE the brain as an information-processing device,**
 - can be traced back at least to the works of William James (1842–1910). Helmholtz also
 - insisted that perception involved a form of unconscious logical inference.

Psychology (1879-)

- ❑ Scientific methods for studying human vision
 - ❑ Hermann von Helmholtz (1821-1894), Wilhelm Wundt (1832-1920)
- ❑ Introspective experimental psychology
 - ❑ Wundt
 - ❑ Results were biased to follow hypotheses
- ❑ Behaviorism (prevailed 1920-1960)
 - ❑ John Watson (1878-1958), Edward Lee Thorndyke (1874-1949)
 - ❑ Against introspection
 - ❑ Stimulus-response studies
 - ❑ Rejected knowledge, beliefs, goals, reasoning steps

Psychology

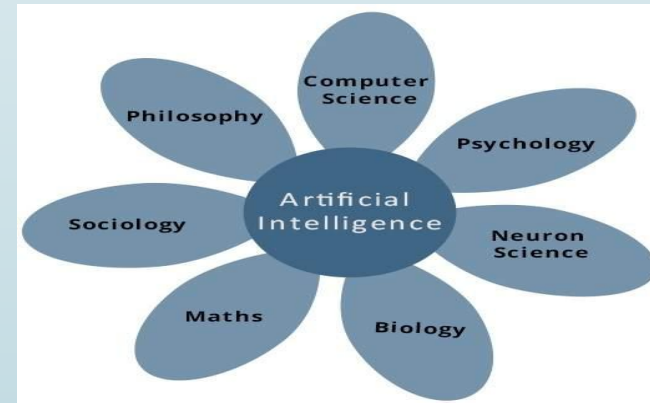
- Cognitive psychology
 - Brain poseses and processes information
 - Kenneth Craik 1943: knowledge-based agent:
 - Stimulus -> representation
 - Representation is manipulated to derive new representations
 - These are translated back into actions
 - Widely accepted now
 - Anderson 1980: “A cognitive theory should be like a computer program”

THE FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

□ the disciplines that contributed ideas, view points, and techniques to AI.

□ **1.2.1 Philosophy**

1. Can formal rules be used to draw valid conclusions?
2. How does the mind arise from a physical brain?
3. Where does knowledge come from?
4. How does knowledge lead to action?



1.2.6 Computer engineering

- How can we build an efficient computer?
- For artificial intelligence to succeed, we need two things: intelligence and an artifact. The
- computer has been the artifact of choice.

- each generation of computer hardware has brought an increase in speed and capacity and a decrease in price.

Performance doubled every 18 months or so until around 2005, when power dissipation problems led manufacturers to start multiplying the number of CPU cores rather than the clock speed.



Computer engineering

- *Abacus* (7000 years old)
- *Pascaline*: mechanical adder & subtractor (Pascal; mid 1600's)
 - Leibniz added multiplication, 1694
- *Analytic Engine*: universal computation; never completed (ideas: addressable memory, stored programs, conditional jumps)
 - Charles Babbage (1792-1871), Ada Lovelace

Computer engineering...

[See Wired magazine late Fall 1999]

- *Heath Robinson*: digital electronic computer for cracking codes
 - Alan Turing 1940, England
- *Z-3*: first programmable computer
 - Konrad Zuse 1941, Germany
- *ABC*: first electronic computer
 - John Atanasoff 1940-42, US
- *ENIAC*: first general-purpose, electronic, digital computer
 - John Mauchly & John Eckert

1.2.7 Control theory and cybernetics

□ How can artifacts operate under their own control?

- Ktesibios of Alexandria (c. 250 B.C.) built the first self-controlling machine: a water clock
- with a regulator that maintained a constant flow rate. This invention changed the definition of what an artifact could do.

□ 1.2.8 Linguistics

- • How does language relate to thought?
- In 1957, B. F. Skinner published *Verbal Behavior*. This was a *comprehensive, detailed account* of the behaviorist approach to language learning, written by the foremost expert in the field.
- Modern linguistics and AI, then, were “born” at about the same time, and grew up together, intersecting in a hybrid field COMPUTATIONAL called **computational linguistics or natural language processing**.
- **The problem of understanding language soon turned out to be considerably more** complex than it seemed in 1957.



The Structure of Intelligent Agents

- Agent's structure can be viewed as –
- Agent = Architecture + Agent Program
- Architecture = the machinery that an agent executes on.
- Agent Program = an implementation of an agent function.

Agent programs

- Take the current percept as input from the sensors and return an action to the actuators.
- The agent program takes just the current percept as input because nothing more is available from the environment; if the agent's actions need to depend on the entire percept sequence, the agent will have to remember the percepts.

The TABLE-DRIVEN-AGENT program is invoked for each new percept and

returns an action each time. It retains the complete percept sequence in memory.

- **function** TABLE-DRIVEN-AGENT(percept) **returns an action**
- **persistent: percepts, a sequence, initially empty**
- **table, a table of actions, indexed by percept sequences, initially fully specified**
 - append percept to the end of percepts**
 - action $\leftarrow$ LOOKUP(percepts, table)**
 - return action**

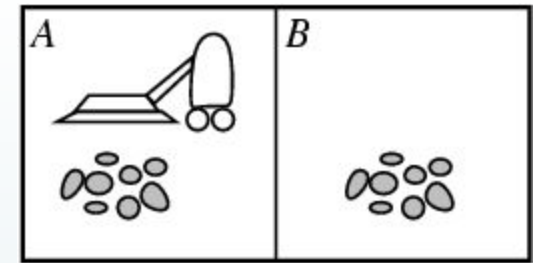
Toy example: Vacuum world.

Percepts: robot senses its **location** and “**cleanliness.**”

So, **location** and **contents**, e.g., [A, Dirty], [B, Clean].

With 2 locations, we get **4 different possible sensor inputs.**

Actions: *Left, Right, Suck, NoOp*

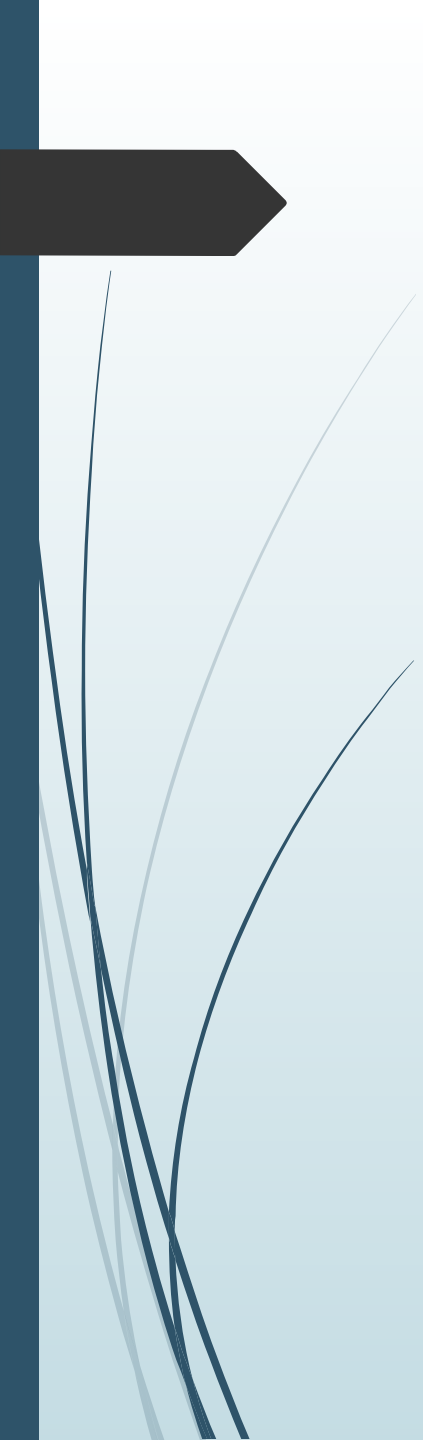


Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

we describe the part-picking robot as episodic, because it normally considers each part in isolation.

- ❖ Several of the answers in the table depend on how the task environment is defined.
- ❖ We have listed the medical-diagnosis task as single-agent because the disease process in a patient is not profitably modeled as an agent; but a medical-diagnosis system might also have to deal with recalcitrant patients and skeptical staff, so the environment could have a multiagent aspect.

- 
- The code repository also includes an **environment generator for each environment** class that selects particular environments (with certain likelihoods) in which to run the agent.
 - For example, the vacuum environment generator initializes the dirt pattern and agent location

States: Beyond Reflexes

- Recall the **agent function** that maps from percept histories to actions:

$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

- An agent program can implement an agent function by maintaining an **internal state**.
- The internal state can contain information about the state of the external environment.
- The state depends on the history of percepts and on the history of actions taken:

$$[f: \mathcal{P}^*, \mathcal{A}^* \rightarrow \mathcal{S} \rightarrow \mathcal{A}] \text{ where } \mathcal{S} \text{ is the set of states.}$$

- If each internal state includes all information relevant to information making, the state space is **Markovian**.

States and Memory: Game Theory

- If each state includes the information about the percepts and actions that led to it, the state space has **perfect recall**.
- **Perfect Information** = Perfect Recall + Full Observability + Deterministic Actions.